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Workpackage WPB

Implementation – Online Job Vacancies

Methodological framework for processing online job adverts data for Official Statistics V.2.

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Purpose and scope

The purpose of this reference methodological framework is to describe at the conceptual level complete statistical online job advertisements data (OJA) processing pipeline according to the four main phases: data ingestion, data pre-processing, information extraction and analysis, and production of statistical indicators. It addresses key conceptual, methodological, technical models and organisational principles and guidelines in using OJA data in official statistics. The idea is that such a framework would be used by statistical agencies when producing statistics using OJA, designing common ESS OJA data processing system and as a basis for future more formal guidelines.

The framework clarifies how OJA data can be used to go beyond the frontiers of conventional approaches of traditional job vacancy statistics (JVS) and identify other possible statistics beyond JVS, which could be produced from OJA data. Traditional statistical approaches, based essentially on surveys, have important caveats: cost, timeliness, accuracy, usage, integration and coverage. OJA data seem to have potential to supplement and enrich traditional statistical data and allow more granular, space-related insights in almost real time, as well as predictive analysis.

Adherence to common standards increases the consistency across the ESS, facilitates sharing and comparing methodologies, services, statistics and data. It also eases software maintainability and identification of risks and potential failures. The aim of this framework is therefore to:

- Provide common language for various stakeholders
- Encourage adherence to common standards, specifications, methodologies
- Provide consistent methods for implementation
- Illustrate various implementation scenarios by identifying common components, processes and systems
- Provide a reference for statistical agencies and other parties to understand, discuss and compare OJA processing systems

Methodological considerations of this framework take into account the developments and conclusions of successfully concluded ESSnet Big data I [1]. Along with the production of relevant methodologies, recommendations and specifications, it was stated that current methods for producing robust job vacancies statistics based exclusively on OJA data were not sufficiently mature [2]. OJA data most likely cannot replace existing job vacancy statistics required by EU regulation. On the other hand, OJA data can provide more granular insights than traditional estimates usually offer. The role of OJA data within official statistics is more likely to provide a basis for producing supplementary indicators.

As far as technical architecture is concerned, this framework proposes the development of a convergence layers system that decouples the collection and pre-processing of alternative data from the statistical concepts. This framework also discuss the role of potential common ESS OJA data collection and processing system as an implementation solution of layered concept (for the time being, based on the OJA system developed by CEDEFOP[1]). Such an approach enables NSIs to focus on remaining statistical challenges and further methodological development. This document introduces general concepts of a layered system and defines roles and responsibilities of the main stakeholders.

[1] European Centre for the Development of Vocational Training

Various scenarios for data governance, management and integration – all of them being compliant with statistical regulatory requirements - are proposed.

The document contains three parts. The first part, **traditional labour market indicators and concepts**, describes traditional labour market concepts. The second part, **methodological considerations around OJAs**, deals with quantitative and qualitative description and analysis of the web data source and provides a few probabilistic approaches on how to integrate and link the web data source with in-house data, i.e. develop new statistics around web data, which can complement statistics from the jobs survey. The last part, **architectural considerations around OJAs**, contains a thorough description on a possible sustainable business process capable on integrating web data into official statistics production systems.

The target readers and users are statisticians, researchers, developers, methodologists developing OJA production systems and statistical indicators in the frame of online job adverts data.

Traditional Labour market indicators and concepts

The rapid technology development in recent decades and digitalization of all life areas, the automatization of a number of processes in production and services, as well as the use of artificial intelligence and the growing needs of the consumers are the main prerequisites for the acceleration of the process of emerging of new professions and significant changes in the existing ones. These processes also predetermine the necessity of new knowledge and skills of the labour force. After the world economic crisis of 2008-2011, the national economies of the European Union countries started growing, which in turn led to growth in labour force demand. At the moment there is lack of labour force in many of the countries, especially in professions that require a high level of technological skills. This situation is also aggravated by the processes of population aging and the migration of highly educated young people. The development of competitive and innovative European economy is based on the need for qualified labour force with specific knowledge and skills. In this regard in June 2016 the European Commission adopted a new agenda for development of skills – New Skills, related to the above stated political priority – a new boost for jobs, growth and investments. The goal of the agenda is to ensure that the right training, skills and support will be available to the EU citizens, so that they could be equipped for quality jobs and realize their potential of confident and active citizens. This will raise the aptitude for employment, competitiveness and growth in Europe (https://ec.europa.eu/social/main.jsp?catId=1223).

The up-to-date and timely statistical information on the availability of job vacancies and on the requirements for specific skills for taking the jobs is a key element in preparing adequate policies for employment, education, including continuing education, retraining and career development at European, national, regional and personal levels. For almost two decades the job demand and supply has been carried out via the Internet. Daily job vacancies ads are published on specialized portals, companies' websites, social networks and mobile apps, and potential workers and employees submit their CVs. The huge volume of information for the demand and supply of labour on the Internet generates the so-called big data, which together with data of traditional surveys could be used for statistical study of this phenomenon and providing real-time data to the labour market.

The economic uplift in recent years, which is a characteristic feature for most of the EU countries, led to unemployment reduction. This in turn reflected in the rise of demand and the use of all sorts of recruitment channels. In active use are specialized web portals, ads on the companies' websites, social media and recruitment agencies. Using data of these online ads has a number of advantages. Furthermore, the growing computer literacy of the population predetermines the Internet as the main channel for searching and finding the right workplace. The big data for the job vacancies generated on the Internet could be used successfully for the needs of official statistics for complementing traditional sources for job vacancies statistics. Their main advantages are their relevance, timeliness, comparatively low costs for their producing and lack of or minimal burden for the respondents. Being unstructured, most online job ads contain information not only for the vacant position but also for the required qualifications and skills, including the so-called soft skills, for degree and kind of the graduated education and professional experience, necessary for the potential applicants for taking it. Often, they also contain information on the employer (sector of the economy, size of the enterprise, kind of ownership, etc.). By web scraping of on-line job advertisements (OJAs) estimates could be made for labour demand by economy sectors, professions, specific skills and qualifications, and regions.

However, using big data as an alternative information source for statistical surveys in any case is accompanied with a lot of challenges. In general, they are related to the difficulty in defining the statistical unit, with the incomplete coverage of the target population and difficulty to evaluate what part of the target population is covered, as well as to what extent certain subpopulations are covered more than others [32]. When using big data for studying the job vacancies the challenges are specifically related to the following problems:

- Defining the scope of the study. The OJAs refer only to part of the actual job vacancies and not all workplaces are announced online regardless of the kind of channels. Some positions are more likely to be published on the Internet and the received estimates will contain deviations connected with the occupations and qualification. As a rule, workplaces are announced on the Internet to a much greater extent in the high-tech economy sectors, as well as in such requiring higher education degree and specific qualification.
- There is unique and reliable information source for job vacancy ads and sometimes they are published simultaneously on different Internet channels – on specialized websites for publishing job vacancy ads, on recruitment agencies' websites, enterprises' websites, on social media, etc. Often some websites republish ads from other ones. This raises the issue of constructing methods for avoiding the so-called duplication of one and the same ad in one and the same Internet source or in different Internet sources and solving of this same issue is among the main challenges for using big data for monitoring job vacancies.
- OJAs often contain both structured and unstructured elements. In the texts of the ads in unstructured form are the usually required education, qualification and skills, which should be turned into structured elements through special procedures. This requires using of specific classification algorithms adjusted for the respective language, which is used by the published ad. However, the data received through these algorithms contain in any case inaccuracies and errors.
- In the various ads the information about the vacancy is available with different extents of details. Most often, the ads of the recruitment agencies do not contain employer data. This leads to some difficulties of collecting data for the industry sector, as well as of directly connecting OJAs data to the official statistical data. Although it's possible to infer the industry sector of the on-line job position through machine learning, the classification errors could occur when processing unstructured data.

All these challenges necessitate careful searching for ways and approaches for estimating the job vacancies via information from online ads.

This document is aimed at revealing the opportunities for using the big data in statistical surveys on the job vacancies in the EU, as well as highlighting the advantages and challenges of this approach for collecting data for complementing the information collected with the traditional statistical methods in this area.

Labour market main concepts

The labour market is usually described and seen in terms of balance between supply and demand in labour force. In public statistics, data on labour supply come mainly from the Labour Force Survey (LFS) approach, while the need of data on demand is fulfilled by surveys conducted in the universe of business or enterprises. Although the Job Vacancy Statistics (JVS) are included in the Eurostat LAMAS

working groups, the JVS belong from a practical point of view to the realm of business statistics. Table 1 summarizes the main concepts of both approaches of the labour market, supply and demand in labour force.

Table 1: Labour supply versus labour demand

	Definition	Main sources
Labour supply (LS)	Workforce supplied by the physical persons to the economy LS= Employed+unemployed persons (according to ILO concepts) Coverage: NACE 1-99 Threshold: 1 hour worked/week	Population registers or labour force surveys
Labour demand (LD)	Workforce the economy demands to the physical persons Inland concept; Coverage: NACE 5-98 LD=occupied posts + vacant posts Threshold: variable (from 1 to 10 employees) Vacancy rate= vacant posts / LD	Business registers or business surveys

Job adverts main concepts used

As discussed in the previous work by the Big Data ESSnet WP1[2], there are fundamental methodological issues around the quality of online job advertisements (OJAs). These issues need to be analysed in order to use these data to produce supplementary indicators for official statistics In particular, the main ones regard what the OJAs represent and cover and how they can be compared with official estimates produced from the official job vacancy survey.

The job vacancy advertisements on job portals (OJAs) cannot be considered similar to the job vacancies (JVs) as defined in the EU Regulations[3]. There are differences between the measures which derive from the definition used by the European official surveys and what can be practically measured from on-line job advertisements.

In particular, official job vacancy statistics, regularly transmitted to Eurostat by Member States and also published in the Eurostat online datawarehouse, are subject to three different EC Regulations.

Official job vacancy statistics provide information on unmet labour demand. In particular, quarterly data on job vacancies are used by the Commission (the Directorate-General for Employment, Social Affairs and Inclusion and the Directorate-General for Economic and Financial Affairs) and the European

[2] Web Scraping for Job Vacancy Statistics in the SGA-2 (that ran from August 2017 to the end of May 2018) and in the SGA-1 (February 2016 to July 2017).

[3]Job vacancy statistics (JVS) are collected under the framework regulation and two implementing regulations: the Regulation (EC) No 453/2008 of the European Parliament and of the Council of the 23 April 2008 [11] on quarterly statistics on Community job vacancies; the Commission Regulation (EC) No 1062/2008 of 28 October 2008 implementing Regulation (EC) n. 453/2008 of the European Parliament and of the Council on quarterly statistics on Community job vacancies, as regards seasonal adjustment procedures and quality reports; and the Commission Regulation (EC) No 19/2009 of 13 January 2009 Implementing Regulation (EC) n. 453/2008 of the European Parliament and of the Council on quarterly statistics on Community job vacancies, as regards the definition of a job vacancy, the reference dates for data collection, data transmission specifications and feasibility studies.

Central Bank to monitor short-term developments in the business cycle and the labour market. The job vacancy rate is one of the principal European economic indicators (PEEIs), the primary source of information used to analyse and monitor short-term cyclical economic developments within the EU, the euro area and individual EU countries. JVs are also used for indicator-based structural analyses conducted as part of the Europe 2020 strategy for smart, sustainable and inclusive growth. The job vacancy rate is also included in the employment performance monitor, a selection of key indicators in the labour market that are monitored in the context of the Commission's Joint Employment Report (JER).

On the basis of the EC Regulation No 453/2008 a job vacancy is defined as:

"*a paid post that is newly created, unoccupied, or about to become vacant:*

a) for which the employer is taking active steps and is prepared to take further steps to find a suitable candidate from outside the enterprise concerned; and

(b) which the employer intends to fill either immediately or within a specific period of time."

The active search for a suitable candidate must have already started but not yet ended at the reference time at which job vacancies are measured. The active steps to find a suitable candidate include: (i) notifying the job vacancy to the public employment services; (ii) contacting a private employment agency/head hunters; (iii) advertising the vacancy in the media (for example the Internet, newspapers, magazines); (iv) advertising the vacancy on a public notice board; (v) approaching, interviewing or selecting possible candidates/potential recruits directly; (vi) approaching employees and/or personal contacts; (vii) using internships.

EC regulation 453/2008 has several mandatory elements:
• Quarterly seasonally adjusted data
• Data broken down economic activity (using NACE)
• Data which are relevant and complete, accurate and comprehensive, timely, coherent, comparable, and readily accessible to users

There are other elements that have been optional until now, or subject to feasibility, including:
• Job vacancies in the agriculture, forestry and fishing sectors
• Job vacancies in public administration, defence and education
• Data on businesses with fewer than 10 employees
• Distinguishing between fixed term and permanent jobs
However, the task force that has been currently working on the review of the legislation in the field of Labour Market Indicators (LMI) will define as mandatory also the extension of JVs data to the public sector and small enterprises with fewer than 10 employees.

The above definition of a job vacancy in the EU regulation framework makes it clear that the concept of OJAs does not correspond to that of JVS, due to different factors.

In particular, the online job vacancies should cover all those job vacancies, as defined by the EU Regulation, for which the active steps carried out by the employers to find a suitable candidate include also the advertising on Internet job portals. But, not all job vacancies are advertised on-line: although there is a general trend towards more job vacancies being advertised on-line, many continue to be

filled through traditional channels, such as newspapers, employment agencies, noticeboards, or personal contacts. The results of the Slovenian study within the WP1 pilot suggest that only about 40% of all job vacancies in Slovenia are advertised on-line. A similar proportion has been reported for France, although this figure may be higher in other countries and may change significantly with time and after the COVID-19 crisis. Evidence from Switzerland suggests a good accuracy between OJAs and JVS.

Furthermore, job portals used by the employers may not be totally covered by the web scraping and other data ingestion activities. This is due to the fact that there is no definitive source of OJAs. Even if the situation varies between countries, OJAs data are generally characterized by multiple job portals, with different business models and complex interplay between them. Job boards only publish original ads uploaded by employers. Job search engines republish ads from other portals. Hybrid job portals are a combination of both. Some portals advertise many different types of jobs while others may specialize in specific sectors. In addition, new job portals may appear while existing portals may decline in importance. All these factors mean that it is challenging task to capture all jobs advertised on-line.

However, even if all the advertisements published on all online portals used for the job advertisements were considered, it is not clear if all the job advertisements posted on all job portals refer to total job vacancies that actually exists at a specific reference date. A job ad is only a proxy measure for the existence of a job vacancy within a company.

For example, delay in the communication between enterprises and the online portals could cause the presence on portals of job vacancies that are closed (because, for example, a suitable candidate has already been found). This should mean that the stock of vacancies as measured by the OJAs should be higher than the corresponding JVs data as measured by the current official surveys on job vacancies. In addition to this, either some OJAs may not represent a job vacancy in the scope of the official survey (these include non-existent vacancies, referred to as “ghost vacancies”, international jobs, and non-payed student internships) or some OJAs may contain more than one vacancy.

Furthermore, the number of vacancies collected by the official job vacancies survey is the stock of vacancies for which businesses are actively seeking recruits from outside their organization at a specific reference date. Online job ads represent a flow of new vacancies but usually do not contain explicit information on when the recruitment activity started and when it will be concluded.

Finally, ads my be published on the web (as they used to be in newspaper) without any direct correspondence with existing vacancies, for example for attracting new talents as a recruiting strategy, or improving the image of a company. Nowadays, since HR recruitment may be managed using IA tools, such “fictional” ads can be handled at very low cost.

The former considerations contributes to address the question of the correspondence between the OJAs and the JVs as defined by EU regulations and suggest to consider OJAs as proxies for actual JVS.

Short description of the JV survey process

EU job vacancy statistics are compiled on the basis of data provided by the participating countries. Member States are granted considerable flexibility regarding the implementation of Regulation 453/2008 in the national statistical systems. Some countries use stand-alone surveys, while others

combine the job vacancy survey with other business surveys. Although the Regulation states that the data shall be collected using business surveys, the use of administrative data is equally permitted under the condition that the data are “appropriate in terms of quality” (according to the quality criteria of the European Statistical System). All the countries, except Czechia, Luxembourg and Croatia, conduct a survey to estimate the number of job vacancies. Czechia, Luxembourg and Croatia use administrative sources.

The minimum information required is the number of job vacancies and the number of occupied posts. Data on occupied posts are generally obtained from the same survey as data on job vacancies.

Surveys are conducted mainly by using postal and online questionnaires or Computer Assisted Telephone Interviewing (CATI) or Computer Assisted Web Interviewing (CAWI). As mentioned, administrative sources are also used.

The basic statistical unit for the data collection is the establishment or local unit. However, the majority of participating national statistical offices collect the vacancy data from enterprises.

As regards, the statistical population, the data cover all enterprises with one or more employees. In France, Italy and Malta only business units with 10 employees or more are surveyed.

Data on job vacancies and occupied posts represent the stock of vacant and occupied posts at a given reference date or averaged across a given reference period. There is currently no internationally agreed rule for the time of recording of job vacancy statistics. Depending on countries, the time of recording for quarterly job vacancy statistics may be one specific day in the quarter (e.g. the 15th of the middle month, the last calendar or working day of the quarter) or a three months average.

As required by the framework regulation, Member States are required to provide the number of job vacancies and the number of occupied posts that can be considered representative for the reference quarter. The preferred methods to achieve this are data collection on a continuous basis or the calculation of a representative average of the data collected for specific reference dates.

The data cover all economic activities defined by NACE Rev. 2, the common classification system for economic activities, except for the activities of households as employers and the activities of extraterritorial organizations and bodies. Covering agriculture, forestry and fishing activities is optional. Denmark provides data only for NACE sections B-N. In France and Italy, in the case of public administration, education and human health (NACE Rev. 2 sections O, P and Q), public institutions are not covered.

The response rates of the surveys vary between 62% and 99% amongst the different Member States.

The quarterly data are broken down by economic activity (at section level), in accordance with NACE Rev. 2 (Statistical Classification of Economic Activities in the European Community). In addition, Eurostat publishes for some countries the quarterly data broken down by occupation (in accordance with ISCO nomenclature) and region (in accordance with NUTS nomenclature), which are delivered by the countries on a voluntary basis.

Data are required to be seasonally adjusted. It is mandatory to supply seasonally adjusted data on job vacancies and occupied posts at least for individual/aggregated NACE Rev. 2 sections B-E, F, G-I, J, K, L, M-N, O-Q, R-S.

Member States whose number of employees represents more than 3% of the European Union total transmit the aggregate number of vacancies and occupied posts within 45 days after the end of the reference quarter. Eurostat publishes JVS flash for the EU around 5 days after the deadline for the data transmission. Within 70 days after the end of the reference quarter, Member States transmit data broken down by economic activity. Eurostat publishes a news release around 8 days after the deadline for the data transmission.

Peculiarities and weaknesses of traditional Job Vacancy statistics

Job vacancy statistics have been discussed and described in great detail during two Eurostat workshops whose major results are summarized in [12]. They present some unique peculiarities, as well as weaknesses or unsolved questions which are described below

a) **Most units report 0 vacancies:** job vacancies are a very peculiar case among statistical variables in the sense that in surveys most responding units report 0 vacancies, while most vacancies are reported by a few units, mostly by the largest enterprises. For instance, in the Swiss JVS survey for the 2nd quarter 2008, 40'500 of the 47'700 responding units reported zero job vacancies (85%). This share was highest among smaller units (96%), with only 4% of the smallest firms reporting at least one vacancy (class 1 : <5 FTE (FTE=full time job equivalents)), 85% among class 2 ([5-20[FTE), 69% among class 3 ([20-50[FTE) and 47% among the largest local units (class 4 >50 FTE).
b) **Business registers do not record information on job vacancies:** job vacancies do not belong to the standard list of variables of business registers. As a consequence, JV survey designs are therefore based on other kind of information as auxiliary variables for sample drawing and grossing up (e.g. number of jobs, employees, VAT, etc.).
c) **Lack of auxiliary information for the treatment of missing values**: because of the absence of auxiliary information relevant to job vacancies, the treatment of missing values in JV surveys is still an open and often unsolved question.
d) **Job vacancies in the branch "Employment activities" (NACE 78):** job vacancies in the branch "Employment activities" is still an open and debated question. Reliable data on Job vacancies are very difficult to get in this branch and their distribution, either in the NACE 78 or in the paying sector are still under discussion in the LAMAS group.

Methodological considerations around OJAs

OJA web scraping in the context of the big data life cycle

The big data processing lifecycle for the production of official statistics (ESSnet on BD I, IT Report 2018) is almost similar to the Generic Statistical Business Process Model (GSBPM 5.1, 2018). In general, a big data process is considered to be composed of 5 main phases:

Acquisition and recording

Extraction, cleaning and annotation

Integration, aggregation and representation

Modelling and interpretation

Dissemination

The above-mentioned big data life cycle phases are identified as the Collect, Process, Analyse and Disseminate phases in the GSBPM 5.1.

In the first step *Acquisition and recording (GSBPM 4 Collect),,* it is all about getting access to the OJAs through web-scraping technologies to acquire and store OJAs data. Typical for this step is that the data are made available in their 'raw' (unprocessed) form as html text files and need to be processed in the next phase in order to be ready for further analysis.

The process of surveying of the job vacancies from the information published on the Internet begins with a choice of sources. Online portals, Internet websites of companies and social networks could be pointed out as main sources. The online portals are public and private. The public portals are most often created, maintained and managed by public employment agencies at national, regional or local levels. The private portals could be owned and managed by national companies or by international consortia. There are also portals which collect ads published elsewhere. Some recruitment agencies also have their own Internet portals. Besides, many companies, especially the bigger ones, have specialized sections for job vacancy ads on their own Internet portals. There are also such sections on some online media, as well as on portals of any kind. In the social networks all these portals could have their own pages, as well as there could exist specialized pages and groups for publishing job vacancies ads. There exist also search engines of job vacancy crawling the portals and companies with published ads by pre-set criteria. In recent years the so-called hybrid portals became popular. They combine their own portal where the employers could publish ads with a search engine in other portals. Some of them offer the employer's standard interface by which the ad uploaded on the company website is accessible also via the portal.

The next two steps *Extraction, cleaning and annotation and Integration, aggregation and representation (GSBPM 5 Process)* cover everything necessary for processing and preparing the raw

scraped OJAs data prior to analysis. The major concern here is the quality improvement of unstructured OJAs data. Because of that techniques and methods for data cleaning, deduplication, data classification, data adjustments, etc., editing sub-processes are applied.

After the stage of collecting OJAs data it is necessary to carry out preliminary processing of the received information aimed at structuring in the form of a database with statistical information. All sources contain a text for the job vacancies which could not be processed through traditional statistical methods. That is why via different techniques for processing of the text content it should be structured in a proper way for extracting the meaningful statistical insights. At this stage it is necessary to remove the so-called noise, which represents information that is not related to the job vacancies, e.g. ads, presentations of companies and opportunities in drop down menus. If this noise is not removed, the information would be incomprehensible and improper for the aims of the statistical analysis. Also, it is of key importance to avoid the repeated counting of job advertisements between or/and within some portals. As mentioned, often one and the same job vacancy is published on several different places. Therefore, it is necessary that the information be collected and based on pre-set criteria the ads for one and the same workplace to be identified and the duplicates be removed.

Part of the preparation may be combining the cleaned OJAs data with administrative sources, e.g. job vacancies data from the Public Employment Agencies at national level (if they are available). Although the OJAs data may still be unstructured at this stage, as a consequence of the adjustments made, the data become more and more structured. This process is not linear and contains many cycles as findings at the end of the chain initiate the need to improve some, or a number, of the previous steps.

The *Modelling and interpretation (GSBPM 6 Analyse)* step is referred to analysing of processed OJAs data from the previous step by applying the machine learning based approaches, using Python, R, etc. The final goal of this stage is to produce new insights and then to calculate the statistical indicators for OJAs. These indicators can be used as a proxy to measure the job vacancy market at national and EU levels. In the last step *dissemination (GSBPM 7 Disseminate)* it is all about producing statistical outputs and then visualizing them to make the data better understandable for analysts.

Benefits of OJAs data

Developing meaningful indicators based on OJA data

The answer to the question „what do the others do in the world“ provides first ideas about meaningful indicators based on OJA data. Regarding public law institutions worldwide, some of them look back on a long experience with this kind of data. In the United States the member driven think tank The Conference Board uses data collected from online job sources including traditional job boards, corporate boards, and social media sites. Their co-called HWOL index, The Conference Board Help

Wanted Online Index, was first published in July 2005, providing users with a data series of total and new online job ads.

The most recent important development is the so-called real time economic tracker from Opportunity Insights, an American organisation based at Harvard University. This tracker has many different indicators and is more or less an immediate consequence of the COVID-19 pandemic, which has significantly increased the already existing pressure to use alternative new digital data sources. For OJA data they publish weekly the change in weekly unique job postings indexed to January 4-31 2020 with a time lag of 3-5 days after the end of a working week.

One can get further ideas from exploring examples from private companies, which collect and analyse OJAs. The six most famous companies are Burning Glass, Haver Analytics, Gartner Inc., Indeed Hiring Lab, LinkedIn and Textkernel. Most of these companies used the potential of real-time information from job portals and react in a special way to the current extraordinary COVID-19 crisis with special web pages.

All of these indicators are based on the idea of providing timely, short-term indicators that show current and policy-relevant developments with regard to changes in the labour demand. Besides that, OJA data provide also the possibility to make some structural analyses whose results do not need to be published at such a high frequency [14].

One of the most popular structural analyses is that on jobs and skills requested in online job advertisements. This is what CEDEFOP for example does with Skills-OVATE or Skills Panorama. They provide information on occupations, skills and regions based on international classifications. Regarding the Swiss Job Market Monitor there is another structural analysis of interest: the structural changes in the importance of different recruiting channels. Germany developed during this project a labour market concentration index (LMC) based on OJAs in order to measure regional concentration OJAs broken down to Functional Urban Areas (FUA).

For a job portal owner there are many other structural analyses possible, as they have access to much more information than only on OJAs. Many job portals have also data from the jobseekers, like information on job wishes or preferences, employment histories or complete resumes (CVs). They can also examine the search behaviour of jobseekers.

Possible indicators compared to traditional job vacancy statistics

"*….OJV data cannot be used to directly replace the existing job vacancy statistics that are required by EU regulation. Indeed, the quality issues are such that it is not clear if these data could be integrated in a way that would enable them to meet the standards expected of official statistics. On the other hand, OJV data can provide more granular insights that official estimates usually offer. It was concluded that the role of OJV data within official statistics is more likely to be as the basis for producing supplementary indicators.*" (ESSnet Big Data Specific Grant Agreement No 2 -SGA2, Final Technical Report, version 2018-05-30, p. 5).

The official data on job vacancies, as required by the EU Regulations and regularly transmitted to Eurostat by Member States, are collected primarily for the purposes of calculating the job vacancy rate, which is one of the Principal European Economic Indicators (PEEIs) for the labour market monitoring.

The job vacancy rate measures the proportion of total posts that are vacant, according to the definition of job vacancy above, expressed as a percentage as follows:

$$\text{Job Vacancy Rate} = \left(\frac{\text{Number of job vacancies}}{\text{Number of occupied posts} + \text{Number of job vacancies}} \right) \times 100$$

The job vacancy rate is required at national level, disaggregated by economic activity sections, enterprise size and on a quarterly basis.

The official job vacancy statistics are not required by EU Regulation broken down by other characteristics of the vacant job position, such as the geographical area/region where the vacant post is located and the profession required. However, these kinds of characteristics are considered as particularly useful by current and actual users, and even if the production of job vacancy indicators broken down by these two characteristics is not required as obligatory by the current EU Regulations, Eurostat already invites Member States to disseminate and transmit these data, if available. For some Member States these data are already available in the Eurostat online data warehouse. Furthermore, the possibility to make obligatory the production of job vacancy statistics broken down by region and profession once a year, in one of the survey waves, is going to be discussed by the Task-force on the Review of EU legislation in the field of Business Based Labour Market Indicators (LMI), whose works are currently in progress.

Table 2: Comparison of JVS and OJA characteristics

Dimension	**Required information in Official JVS**	**Potential of OJA**
Timeliness	70 days after the reference quarter and 45 days[4]	Real time
Frequency	Quarterly	Daily
Economic activity sector	Yes	Yes
Enterprise size	Not compulsory	Yes
Occupation	Not compulsory	Yes
Skill	No	Yes
Education	No	Yes
Geographical area/region	Not compulsory	Yes
National totals (estimate)	Yes	No

Moreover, additional information on a monthly basis and within the month at daily level, as that deriving from the OJAs, gives useful insights into the dynamics within the quarter of the labour demand expressed by the online recruitment process. In general, despite the representativeness limits

[4] Only for Member States whose number of employees represents more than 3% of the European Union total.

of OJAs, daily information on them provides valuable insights into the vacancy flows during the quarter and not only into the vacancy stocks at a specific date of the quarter. In the current official vacancy surveys job vacancy data are referred to one specific day in the quarter (e.g. the 15th of the middle month, the last calendar or working day of the quarter) or to a representative quarterly average. Therefore, daily OJAs allow observing the total flow of vacancies opened and closed within the quarter and not only those vacancies still opened at a specific date of the quarter.

Furthermore, the breakdown by economic activity, occupation, region, education together with data at a frequency higher than the quarter could provide useful insights into the change of the job positions characteristics that need to be filled by the enterprises.

Amongst the concluding remarks of the previous phases of the project, it had been shown that while "…it *is feasible to produce estimates of on-line job vacancies, there is a big difference between these and the official job vacancy estimates. It is therefore clear that these could never replace the official estimates. Further, the benefits of these estimates for policy making are not clear as they only give a partial (and not easily defined) view of overall labour market demand. It therefore seems that the role of OJV data within official statistics is more likely to be as the basis for producing supplementary indicators. These could include indicators of local labour market demand (as shown above) and/or indicators of occupation groups and associated skills. However, rather than measuring absolute levels, these would be more useful for measuring change over time*…"(ESSnet Big Data Specific Grant Agreement No 2 -SGA2, Final Technical Report, version 2018-05-30, p. 35)

Therefore, detailed *supplementary* indicators could be derived from OJAs.

Table 3: Potential indicators from OJAs

Indicator	**Frequency**	**Grouped by**
Number of available on-line job vacancies in a reference day	daily	economic activity, occupation, region, education
Number of available on-line job vacancies in a reference month	monthly	economic activity, occupation, region, education
Number of newly posted on-line job vacancies in a reference day	daily	economic activity, occupation, region, education
Number of newly posted on-line job vacancies in a reference month	monthly	economic activity, occupation, region, education
Average number of OJAs	weekly , monthly, quarterly	economic activity, occupation, region, education

Percentage changes over the same period of the previous year (year on year changes) of the number of OJAs	daily, weekly, monthly,	economic activity, occupation, region, education
Percentage changes over the same period of the previous quarter (quarter on quarter changes) of the number of OJAs	daily, weekly, monthly,	economic activity, occupation, region, education
Number of available on-line job vacancies at the JVs reference day	quarterly	economic activity, occupation
Daily pseudo-stocks of job ads based on CEDEFOP data	quarterly	industry sector, region or required qualification

OJA seen as proxies to statistical indicators

Like job advertisements in newspapers before the introduction of the formal Eurostat JVS regulation, OJAs can be seen as proxies for actual job vacancies:

- To un reveal and quantify the link between JVS and OJA
- To determine to which extent OJAs are sufficiently congruent with JVS, to be used safely as reliable measurements of job vacancies.

In statistics, a proxy or proxy variable is a variable that is not in itself directly relevant, but that serves in place of an unobservable or immeasurable variable. [10] (Wikipedia https://en.wikipedia.org/wiki/Proxy_(statistics)#cite_note-1). A proxy is an indirect measure of the desired outcome which is itself strongly correlated to that outcome. It is commonly used when direct measures of the outcome are unobservable and/or unavailable. An organization should use a proxy measure when there is little or no data available about the program being implemented, but the outcome the program is designed to influence has an existing and commonly accepted proxy.

Modelling is the way to use proxies in order to produce statistical indicators. Regression is a simple tool to identify correlations and produce reliable predictors to statistical indicators.

Complementation of JVS

OJAs can be used in several ways to complement the official JVS, in particular as a complementary source:

a) as a new set of variables in business registers. Once JVs included in business registers, it would be possible to design surveys more specifically tuned to job vacancies, taking into account the actual distribution of JVs among enterprise sizes and possibly to reduce the burden on enterprises. This would of course imply a full coverage program of all enterprises

b) to fill non-response units and for imputation purposes in general

c) as auxiliary variables during JVS grossing up processes

d) to improve the JVs data collection of NACE 78

The choice of websites for recording JVs and the quality of the linkage between OJAs and register units are a key issue in this context.

The analysis carried out so far have shown some of the potential enrichments that OJAs could offer for the official job vacancy indicators, as required by EU Regulations.

Despite their coverage and representativeness limits, OJAs have the main advantage of providing information on the job positions announced, not only high detailed in terms of the characteristics of job positions, but also at a high frequency level, providing an information on a daily, weekly and monthly basis. Therefore, in addition to offer a potential enrichment for job vacancy official indicator from an informative point of view, OJAs can support the current production of JV official indicator from a methodological side.

Amongst the several ways in which OJAs can be used to complement the current official JVS production, in the following sections we consider three other further uses:

a) to derive evidence on the vacancy flows during the reference quarters of the official surveys;

b) to evaluate the representativeness of the vacancy stocks at the specific reference dates used in the official surveys;

c) to indirectly calculate a monthly basis JV official indicators from the quarterly ones.

Vacancy flow instead of vacancy stock at specific reference dates

OJAs have the advantage of capturing the vacancy flows that is the vacancies opened and closed during a reference period, instead of only the vacancy stock at specific reference dates.

The vacancy stocks takes only into account the vacancies that survived at a specific date, without reporting the movements of vacancies before that date.

However, information on vacancy flows from a survey would require a collection on a continuous basis. The EU Regulations on job vacancy statistics consider this collection method as the preferred one: “MS shall provide data on the number of job vacancies and the number of occupied posts that can be considered representative for the reference quarter. The preferred methods to achieve this are data collection on a continuous basis or the calculation of a representative average of data

collected for specific reference dates" (EC Reg. No 19/2009 implementing the JV frame Regulation EC No 453/2008).

A continuous collection method implies a significant improvement in the collection costs and in the response burden for the enterprises. From the enterprise side in fact, it would imply having a kind of register in which keeping the account of all vacancies opened and closed during the reference period.

OJAs offer a kind of a continuous based information without any additional costs. Starting from the stock of OJAs at the beginning of a reference period, using daily information on OJAs, it can be calculated the vacancy net balance (the differences between vacancies opened and closed) day after day.

$$JV_d - JV_{d-1} = JVO_d - JVC_d \qquad [1]$$

From a JV survey, in general, it can be computed only the vacancy net balance between the reference dates of two different reference periods (for, example, between two different months or quarters).

Therefore, OJAs allows to have the information on the net increase or decrease of vacancies at a very short time period, by day, week, half a month, etc..

Representativeness of the job vacancy stock at specific reference dates

Even if collecting vacancies on a continuous basis is a "best collection practice" the majority of EU Member States use either a representative average of data collected for specific reference dates during the quarter or a single specific reference date (such as the last calendar day of the quarter).

In addition to the vacancy flows information, OJAs daily distribution during the quarter can give useful insights on the representativeness of the vacancy stock at the specific reference dates used in the current job vacancy surveys.

In particular, OJAs can show whether, and to what extent, the single reference date used is representative for the quarter, since the flow of vacancies opened and closed during the quarter cannot be fully captured from a single specific stock. It is especially true during very peculiar period, such as the COVID-19 emergency months, when the single reference dates used in the official surveys fell during the enterprise total lockdown.

Therefore, it is worthwhile analysing the OJAs daily distribution in order to define the best reference dates of the job vacancy official survey collection, or at least, in order to better know the limits of the currently used reference dates.

Figures 1 to 4 shows the Italian daily OJAs distribution during the fourth quarters of 2019. The Italian job vacancy official survey collects vacancies at the last calendar day of the quarter.

The preliminary analysis carried out so far shows the intense flows of vacancies within the month; the different vacancy patterns over each month and the different monthly average numbers of OJAs within the quarter. However, there is no evidence of a clear OJAs daily pattern during each month. Furthermore, the "best capturing dates" during the quarter are difficult to be detected and there is not a strong signal supporting a change from the currently used reference date to another single one date for the Italian vacancy survey. It should be better moving from the last calendar day of the quarter

to an average over several single dates; but the significant negative impact in the implementation costs for the survey and enterprise response burden need to be take into account.

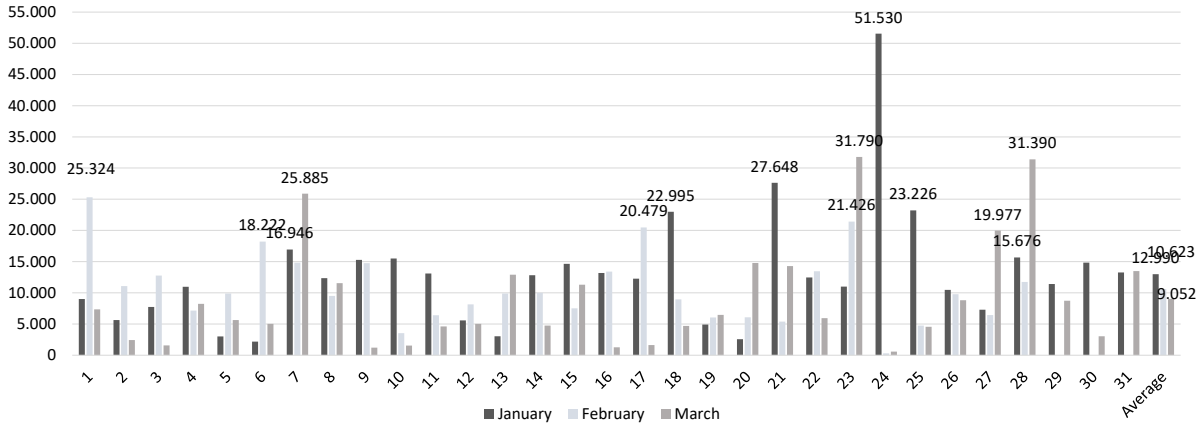


Figure 1: OJAs daily distribution during the four quarters of 2019: the Italian case, first quarter 2019

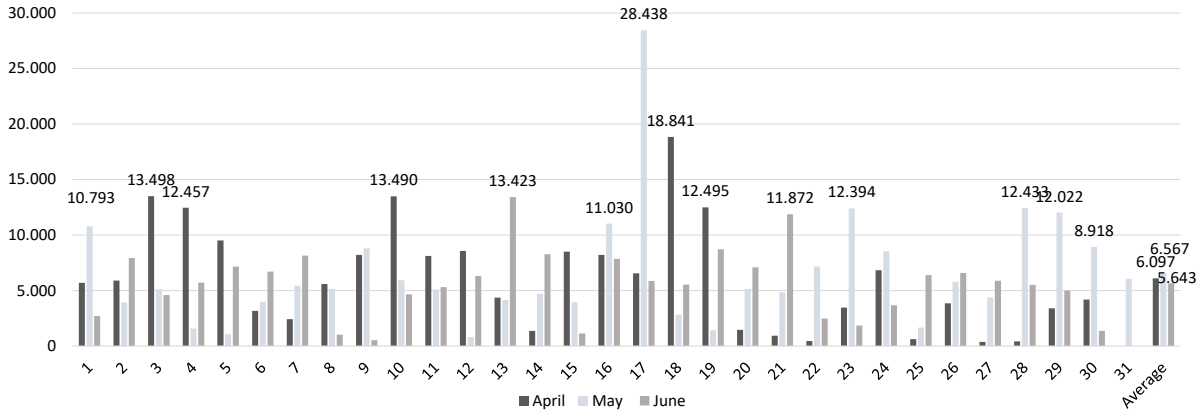


Figure 2: OJAs daily distribution during the four quarters of 2019: the Italian case, second quarter 2019

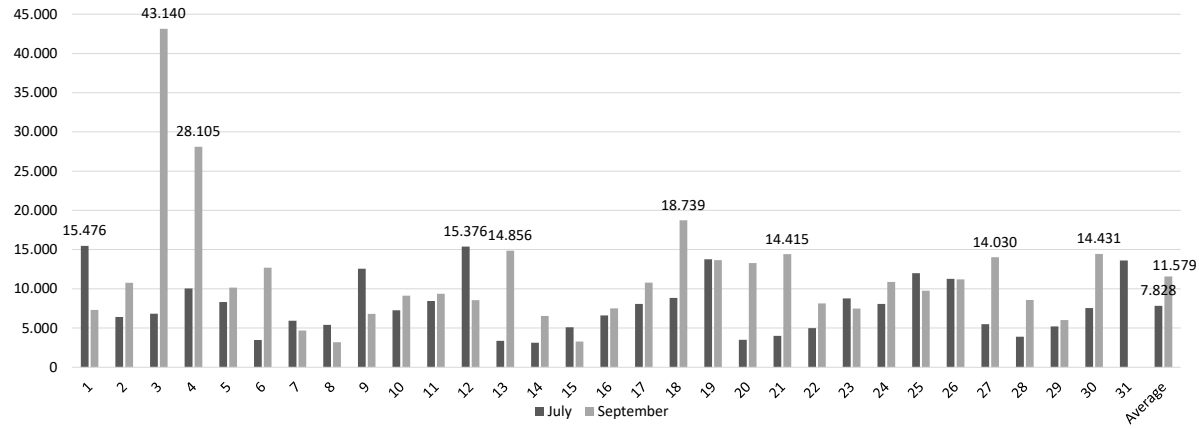


Figure 3: OJAs daily distribution during the four quarters of 2019: the Italian case, third quarter 2019

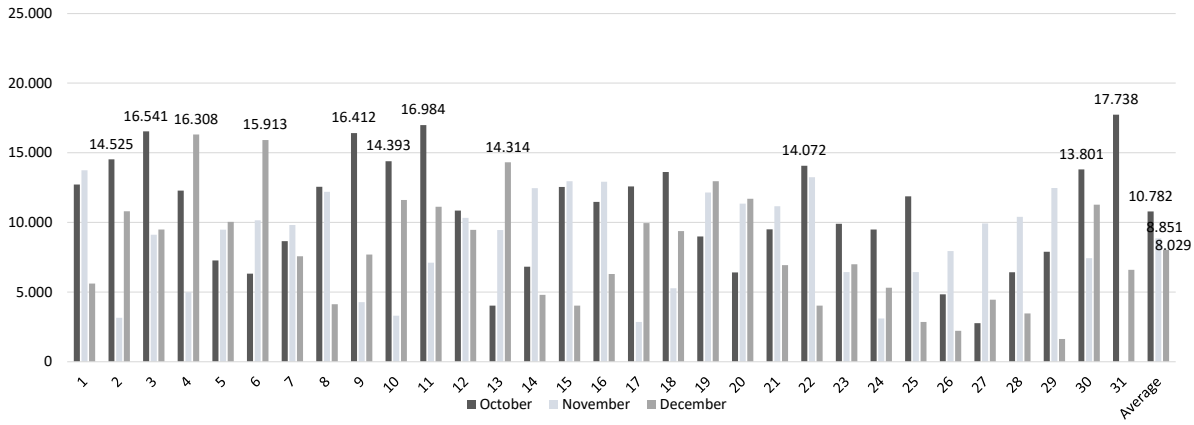


Figure 4: OJAs daily distribution during the four quarters of 2019: the Italian case, fourth quarter 2019

From a quarterly to monthly basis JV indicator

The high frequency detailed information from OJAs can also be used to indirectly extend the frequency of the JV official indicator from a quarterly basis to a monthly one. This is another helpful use of the OJAs, taking into account the difficulties both in implementing a vacancy survey on a continuous basis and in changing the currently used reference date or extending it to more dates.

EU Regulations requires the job vacancy rate on a quarterly basis. In Istat, as already mentioned, for the computation of the vacancy rate, data on job vacancies refer to the last calendar day of the quarter. Daily information on OJAs provides valuable insight into vacancy flows during the quarter. The monthly average number of OJAs could be used to estimate the vacancy rate on a monthly basis, as follows:

$$Monthly\ JV\ rate = quarterly\ JV\ rate * \frac{monthly\ mean\ OJAs}{quarterly\ mean\ OJAs}$$

Since there is no evidence of a clear daily pattern in the months, we preferred to use the monthly and the quarterly average of OJAs instead of the number of OJAs on a specific day (e.g. last day of the month or the quarter). Job vacancy rate is one of the principal European economic indicators (PEEIs), the primary source of information used to analyse and monitor short-term cyclical economic developments within the EU Member States, the euro area and the individual EU countries (ESSnet Big Data II, Methodological framework for processing online job adverts data for Official Statistics V.2, p. 10). Therefore, a monthly job vacancy rate, indirectly derived from the OJAs, can be used to better understand the dynamics of the economic cycle within the quarter, also in relation to other monthly indicators.

A first attempt of estimating a monthly vacancy rate, only for the Italian case, from the third quarter of 2018 to the fourth quarter of 2019 has been carried out. The exercise has been based on the monthly OJAs data over the months July 2018-December 2019 and on the current not seasonally adjusted job vacancy rate quarterly estimates, derived from the Italian official survey (Figure 5).

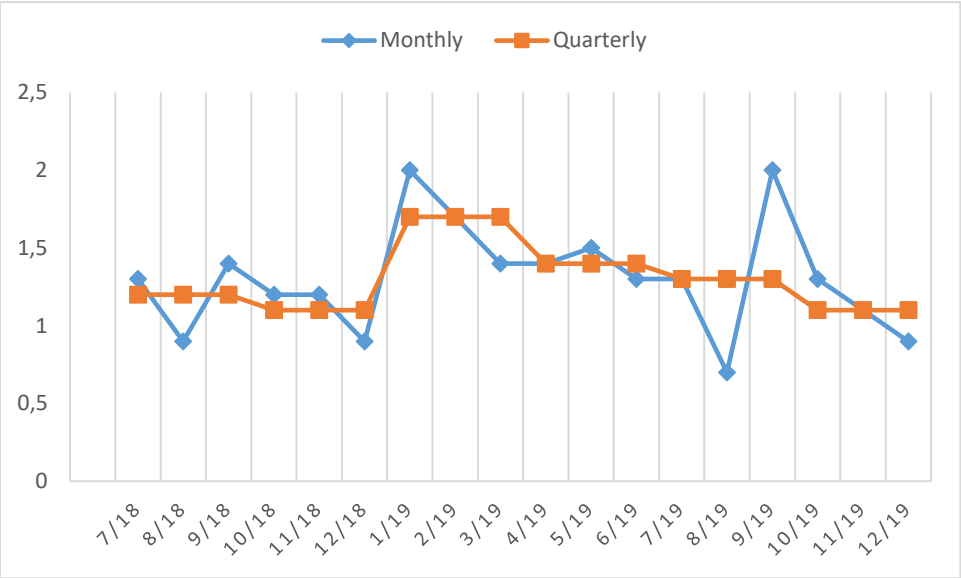


Figure 5: From a quarterly to a monthly basis job vacancies rate estimates: the Italian case

Table 4 shows the vacancy rate year on year changes (percentage points differences), calculated comparing the vacancy rates to the same period, month or quarter, of the previous year. The aim of this table is to compare the monthly dynamics within the quarter with that between quarters and to highlight the possible differences. With respect to the quarterly year on year change, which is an average of the monthly year on year changes, the monthly change gives the additional information of the dynamics inside the quarter. Therefore, it is interesting to note, for example, that the growth in the vacancy rate in the third quarter of 2019 (+0.1) is mainly due to the growth in the vacancy rate in September 2019 with respect to September 2018 (+0.6).

Table 4: Year on year change in the job vacancy rate (percentage points difference): the Italian case

Quarter	Month	Change in monthly JVR	Change in quarterly JVR
Q319	7/19	0	0.1
Q319	8/19	-0.2	0.1
Q319	9/19	0.6	0.1
Q419	10/19	0.1	0
Q419	11/19	-0.1	0
Q419	12/19	0	0

Choice of websites: enterprise websites versus job portals

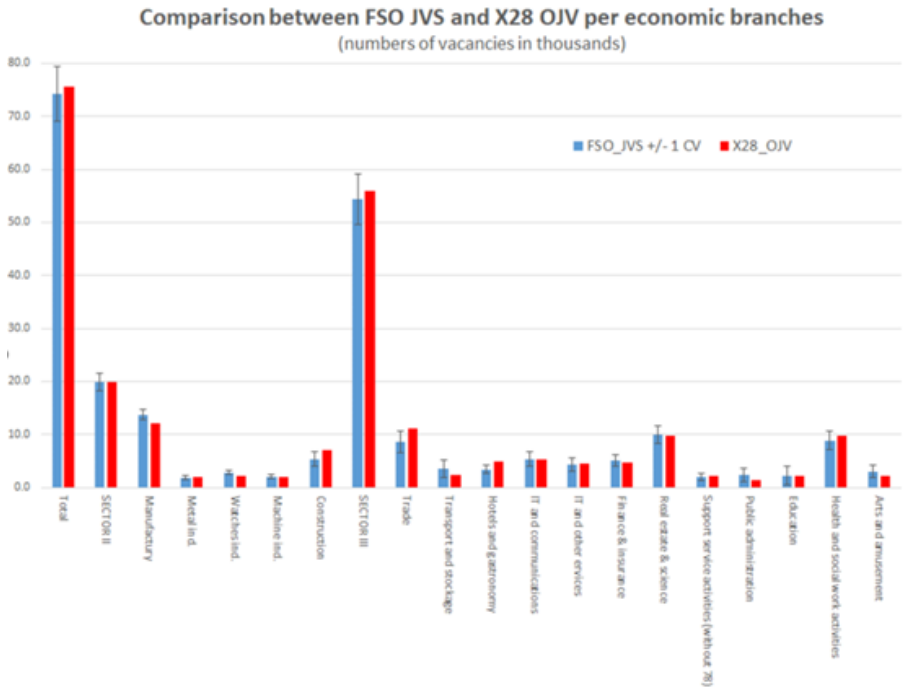
Until now, most attention has been concentrated on job advertisements published by major job portals, which are partly a secondary source of data, since they tend to collect and republish job advertisements from enterprise websites or from other portals. An alternative to this approach would be to collect primary source data, i.e. job advertisements published by enterprises on their own websites.

A good example of such an approach has been conducted for Switzerland comparing data collected by the private company X28 AG with the official JV survey [13]. The analysis of the data for the 4th quarter 2018 shows that a full coverage of primary source data is a realistic alternative to survey data

(Figure 1). Primary source data, i.e. vacancies collected online from websites of enterprises, have been matched to enterprises of the business register. Using the UID as a matching key (https://www.bfs.admin.ch/bfs/en/home/registers/enterprise-register/enterprise-identification/uid-entities.html), a unique match could be found for most units, i.e. 23'600 of 25'200 enterprises, i.e. 93,7%. The remaining 6,3% unmatched units encompassed 3'700 vacancies, i.e. 4.4% of the 85'100 job vacancies of the X28 AG test file. Filtering the matched data for Sections B to S of the NACE and excluding NACE 78 which is not covered by the Swiss JVs statistics resulted in a total amount of 75'600 vacancies from online websites with reference data Dec. 15th 2018 as compared to 74'300 ± 5'100 vacancies for the official JV survey with reference date Dec. 31th 2018. Similarly good agreements were found at NACE1 and NUTS2 levels, with OJAs within the confidence intervals of the JV survey (Figure 6).

More interestingly, a full coverage of all enterprises of the country (ca 600'000) resulted in a OJA dataset of a ca 25'000 enterprises with at least one job vacancy (4.2%), a size similar to the JVs survey (22'000 enterprises; 3.7%).

On the other hand, the total number of online job vacancies recorded by X28 AG, i.e. including vacancies recorded on job portals, and published in the press release Jobradar for the same quarter amounted to 189'900 (https://www.jobagent.ch/resources/jobagent/jobradar_Q4_2018_de.pdf), which indicates that the official number of job vacancies represents only 40% of the total number of advertisements collected online, or a ratio OJAs/JVs of 2.5. This ratio has been increasing since 2013, with an average over the last seven years of 2.4 with a minimum of 1.8 in 1st quarter of 2013 and a maximum of 2.9 in the 4th quarter of 2016.



Legend

Blue bars: JV survey estimates with 95% confidence intervals; red bars: OJAs collected from enterprises websites (primary source data). The first pair of bars from the left corresponds to the numbers of job vacancies for whole Switzerland, the second pair to "Industry & construction: sections B to F," and the eight pair to "Services: section G to S (NACE 78 excluded)".

Figure 6: Comparison between OJAs and JVs for the 4th quarter 2018 in Switzerland

Some quality aspects of OJA data

According to the work done by ESSNet Big Data I, WP8 Methodology, there are five main groups of measures that can be used to assess the data quality in the web scraping process of online job vacancy data: coverage, accuracy and selectivity; measurement errors; comparability over time; linkability and processing errors; model errors and precision. In the current ESSnet Big Data II project, the Quality Guidelines for the Usage of Big Data is being prepared by the WP-K Methodology, where the web scraping is described as a separate big data class.

Coverage and selectivity

As previously discussed, one of the fundamental quality issues in using on-line job advertisement data is that not all job vacancies are advertised on-line. Understanding these issues of coverage has been a key focus of pilot studies carried out in the previous phases of the project, aimed at identifying different approaches for trying to better understand and measure these differences (SGA - 2 Final Technical Report, p 28). In particular, three different approaches had been defined: micro-level comparisons, aggregate comparisons and measuring use of advertising channels via the JVS. The first approach "...involves linking OJAs data to either the reporting units of the JVS or the Business Register. The UK has linked JVS data to vacancy counts by company for a range of OJA sources. This has revealed that the pattern of job counts between the JVS and other sources by reporting unit/company is typically very messy and difficult to understand…". The second one "...involves comparing JVS aggregates by industry sector (i.e. NACE) with the equivalent taxonomies from job portals. This approach has been used by Germany, who does not have access to JVS micro data. While this is quite

a straightforward approach, private portals usually have their own taxonomies, which are often only approximately comparable with NACE…". A similar attempt recently conducted for Switzerland proved to be a very promising approach when a good linkage between OJAs and enterprises can be achieved, with OJAs close to JVS estimates at the economic activity section and regional levels. A third approach "…has involved surveying enterprises and asking specific questions about advertising channels. Surveys of this type have been carried out by both Germany and Slovenia. The German study found that large companies were more likely to advertise on-line, with small companies more likely to use other channels." (SGA - 2 Final Technical Report, p. 28). All these three approaches can be used together in order to better understand the actual coverage differences between OJAs and JVs. By means of these approaches over- and under- coverage issues can be deepened throughout the analysis at the enterprises characteristic level: enterprise size, economic activity sector and geographical area.

As other big data sources, OJAs also suffer from the problem of there not being an appropriate frame for the target population, i.e. there is no list that fully covers the target population. This makes it difficult to define the extent of the over-coverage.

The lack of frame for the target population does not allow calculating the known totals used in the calibration procedure. Therefore, calibrated weights cannot be computed in order to obtain OJA total estimates. As a consequence, OJAs cannot be considered as representative of the total number of vacancies that exists at a specific date.

Furthermore, in big data sources in general, which are based on non-probability samples by definition, the selection bias becomes dominant. In the field of big data sources, the discussion is restricted to biases occurring due to the self-selection mechanism, that is caused by individual sample units, and selectivity is defined as a: "…*general term for self-selection error which results from individuals (unit-specific, e.g. whether to tweet or use a certain mobile phone network provider) and data holder decisions (data holder specific, e.g. in terms of business concept, technical infrastructure). This includes coverage, measurement or non-response (missingness) error, which further introduce potential bias in estimates based on big data sources only*." (see Eurostat, 2018, p. 25)

In particular, the data collected by the web scraping activities suffer from unit-specific selectivity (self-selection) - that is the result of any decision of individuals (or other entities) that affects its presence in the sample (see Eurostat, 2018, p. 25).

In the case of big data in general, there are three types of decisions to take into account:

a) The decision to use the technological platform from which big data are being captured, e.g. the Internet or a device connected to the mobile phone network;

b) The decision to select a particular Internet platform (e.g. one particular social media portal) or a particular mobile operator; and

c) The decision to provide paradata that might be used to impute (pseudo-)responses or auxiliary variables.

In the case of OJAs, in particular, different types of enterprises can use in different ways the advertising on job portals. For example, very small enterprises may prefer to make use of more informal active

steps, such as word of mouth. In addition to this, employers could consider that the advertising on job portals could be more effective for some kinds of job positions, with reference to their characteristics, such as kind of profession required, geographical area, level of education required, etc. Thus, some types of jobs might more likely be advertised on-line than others. This means that OJAs data might not only miss many jobs, but might also not be representative of the overall job market.

Measurement errors

One of the main types of measurement errors of web scraping on job portals are result of scraping errors, i.e. the scraper may construct such as to download incorrect data from the job portals. The other type of measurement error could be the errors on the job portals or employers may upload incorrect data on the job portals. The measured concepts by scraping job portals may not be identical to the official statistical ones required by the NSI and this also may cause measurement errors. Given the nature of these data resembling a flow, some thought has to be given to network connectivity problems, which might disrupt the collection of data massively.

Further type of measurement errors, could be those deriving from classifying online job vacancies by NACE economic activity section and occupation. However, these kind of errors should be better included amongst, respectively, linkability and processing errors. As showed in the next sections, a wrong link between the names of enterprises as they appear from the online job advertisement and the Statistical Business Register (SBR) could produce a wrong OJA classification by NACE economic activity section. Similarly, errors in the processing steps of the raw scraped data from job advertisements could generate a wrong OJA classification by occupation.

Linkability

When scraping OJAs, the goal is to find the number of ads for each company through information from the Internet, so we have to link names (as they appear on the Internet) to official company information in the Statistical Business Register (SBR). At the same time scraped job titles and descriptions do not usually match the official ones and classification codes are not included, so we classify them through machine learning - classification method and then to the official names/titles. In these cases, NACE errors can be occured in the sense that linking to erroneous enterprises may produce erroneous NACE.

The linking of on-line job ads with JVS micro data or SBR data is intended to understand coverage issues of scraped OJAs data. Usually, this is a problematic issue, because the employers use abbreviated names, trading names rather than the legal enterprise name, and misalignment between the company names and the JVS reporting unit.

Linking at the enterprise level between the National Employment Agencies data and the Business Register is straightforward if they both use a common enterprise numerical identifier.

A major issue in trying to match JVS reporting units to company names in job portals is that many jobs are advertised through private employment agencies and the employer is not usually mentioned in the advertisement.

However, the analysis of data of X28 AG in Switzerland indicates that when sufficient energy is put into the linking process very accurate figures can be produced using OJAs from enterprise websites data.

Quality of individual matching in the Swiss example

However, in the example described above for Switzerland, the match is very poor, when cases are examined individually, with a very wide scatter of JVs versus OJAs, while one would expect data aligned on a straight line of slope 1 (Figure 7 - horizontal axis: OJAs collected by X28 AG from enterprise websites (log scale); vertical axis: JVs collected in the survey (log scale); N =3511 units common to the two datasets). How can this be explained? There at least two possible cases :

a) the data collected online and the survey differ widely
b) the matching is incorrect

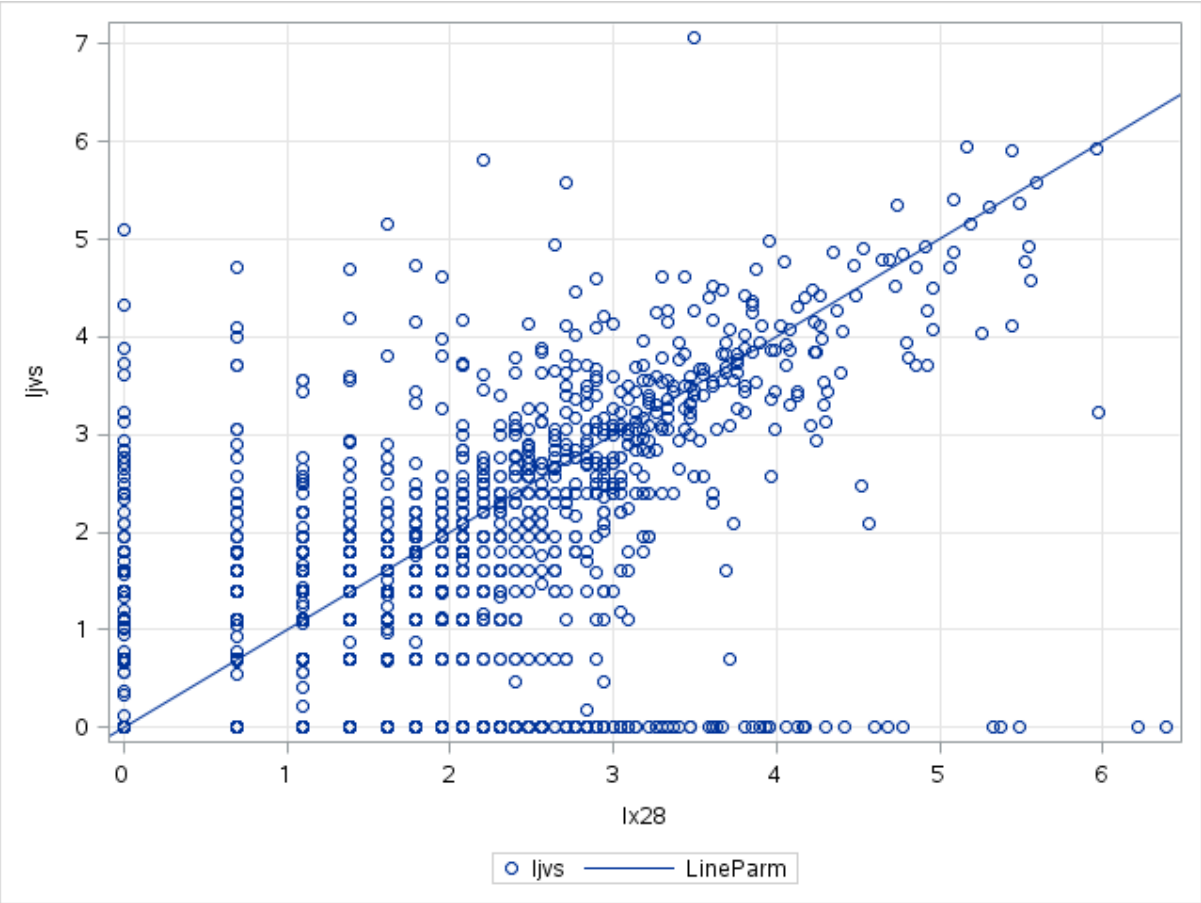


Figure 7: Scatter plot of OJAs common to X28 AG JVs datasets

A close examination of the data indicates that the matching is problematic, because of the complexity of the structure of big companies and groups of enterprises.

Identifying enterprises in business registers

The structure of enterprises in business registers may reach a very high complexity, especially in large enterprises with many establishments and local units. Four kinds of identifiers can be distinguished, each of these categories forming a particular layer:

a) legal units (bear legal attributes with name and address)
b) administrative units (used for administrative purposes with name and address)
c) physical units (bear employees, buildings with localisation, produces goods and services)

d) statistical units (level to which statistics must be produced, e.g. local or kind of activity units)

The structure of enterprises may be also be very dynamic with frequent and profound changes. In cases of closures and re openings, fusions and acquisitions, splittings, even if the physical structure may be little affected (for instance a restaurant whose owner changes may keep the same name, address menu and employees), the legal and administrative units will change, as will the statistical unit. It is no more the same "enterprise". Only specialists of business registers in NSIs are able to properly match data collected from the reporting units to the correct statistical units. Without this important matching effort and knowledge of the enterprises structures, there is little hope to produce statistical products which respect JVS quality standards using OJAs.

Pseudo-stocks

Germany proposed to approach JVs using the concept of "pseudo-stocks". "The idea behind pseudo-stocks is the following: At a given point in time (T), we consider a job ad to be valid if it hasn't expired yet and was posted a maximum of 30 days before T. Expiration of a job ad is determined by the job advert's expiration date. Preliminary analysis of other sources indicates that job ads stay relevant for roughly 30 days on average (see ESSnet WP 1). The number of all job ads which are considered at time T is the pseudo-stock of active job ads at time T". If the results seem to provide pseudo-stocks relatively close to the JVs for the total for whole Germany for the three quarters analysed (3rd quarter 2018 to 1sr quarter 2019), the agreement is not so close when the data are displayed per economic sections, which might result from NACE classification errors.

The same idea had been previously applied to the 13 countries for which CEDEFOP data are available with slightly more complex selection criteria [15]. When the OJAs figures differed too much from the JVs, a closer agreement was searched playing with the time window. The results are shown on Table 5. They indicate rather diverging results among countries with OJAs almost 50% higher than JVs for UK and more than 80% lower for CZ. Extending the time window to 90 days for OJAs did not allow to reach the JVs level for BE, while this level was still 30% higher with a 15 days window for Ireland. There were no JVS data available for IT and FR. This track does not seem very promising unless new criteria are developed.

Table 5: Attempt to extract pseudo-stocks for the 13 countries with CEDEFOP OJA data

		JVS				OJV			Time	atio/OJV_JVS			
		2018Q3	2018Q4	2019Q1		2018Q3	2018Q4	2019Q1	Window	2018Q3	2018Q4	2019Q1	average
AT	Austria	126'335	130'869	128'211		159'253	108'497	84'430	30 days	126%	83%	66%	92%
BE	Belgium	149'180	141'730	147'698		108'292	111'870	146'790	90 days	73%	79%	99%	84%
CZ	Czechia	270'588	277'441	295'904		52'482	47'536	45'221	30 days	19%	17%	15%	17%
DE	Germany	1'227'600	1'445'897	1'366'915		1'517'218	1'732'571	1'365'818	30 days	124%	120%	100%	114%
ES	Spain	124'747	123'772	131'531		112'457	135'822	135'773	30 days	90%	110%	103%	101%
FR	France	:	:	:									
IE	Ireland	19'200	17'400	19'500		20'123	24'565	27'611	15 days	105%	141%	142%	129%
IT	Italy	:	:	:									
LU	Luxembourg	7'527	7'295	7'272		8'076	8'906	8'008	60 days	107%	122%	110%	113%
NL	Netherlands	255'843	252'782	276'107		500'000	322'338	113'765	30 days	195%	128%	41%	121%
PL	Poland	156'303	138'434	141'906		255'760	108'430	96'574	30 days	164%	78%	68%	103%
SE	Sweden	103'006	103'250	142'523		93'560	107'216	166'167	30 days	91%	104%	117%	104%
UK	United Kingdom	865'000	854'000	820'000		1'643'401	1'391'481	731'965	30 days	190%	163%	89%	147%

Processing errors

When dealing with semi-structured and unstructured descriptions of OJAs, we are dealing with mostly text data, so some form of language processing (machine-learning methods) is needed. By definition

this means that either data are pre-processed into features which are substitutes for the true information stored in the text or a form of machine learning is used. Either way, algorithm validity errors should be considered.

The raw scraped data from job advertisements usually require a lot of processing steps before they can be analysed. For example, job title fields often contain additional information, such as job location, occupation (job title), key skills, and salary. This could generate the processing errors, e.g errors in classification of occupation.

De-duplication seemed to be another major issue during the processing phase. There is a need to design specialized software tools or methods for effective removal of the duplicate records in/between sources.

Data integration errors can happen when integrating OJA data from several different job portals or when linked with the data from the JVS or SBR data.

All these reasons suggest that OJAs should not be considered to encompass the total number of job vacancies that exists at a specific reference date.

Fictional OJAs

In some cases, advertisements may be published without actual posts to fill, for example to collect CVs or for marketing or communication purposes. Such cases are virtually impossible to detect.

Detecting duplicated OJA

Ensuring the high quality statistics using OJA data raises many technical challenges. One of them is the detection of duplicates of job ads. Companies advertise their jobs as widely as possible to attract a large amount of good applicants: on corporate sites, on general and specialised job boards, via staffing agencies, etc.

The same job vacancy can be published to different web sites and in different times by the same employer, therefore the duplicate detection is a paramount process to obtain meaningful data (about 30% of the downloaded vacancies are dropped during the duplicate detection process). The vacancy title, description, and publication dates are considered to identify duplicates. A similarity among texts and overlapping or very close publication time is considered as a signal of duplication. Two job vacancies are compared in the following way: stop words are removed, then texts are compared using measures that quantify how dissimilar two strings are to one another by counting the minimum number of operations required to transform one string into the other. When similar texts are found, vacancies are not immediately regarded as duplicates, but the publishing dates are considered. If the dates are overlapping or contiguous, the similar-text-vacancies are recognized as duplicates, while if (at least) a 1-month gap exists, they are not regarded as duplicates. These criteria have been identified and validated in cooperation with several organizations managing some web sites selected for the project.

Analysis of CEDEFOP data for France between sources and sites for French job adverts showed that the majority of sources are sites. Except for Adzuna which seems to only repost job adverts from other sites. Top 10 sources/sites seem to also repost offers from other sites. So most job boards seems to combine proper vacancies and job ads originating from other websites. In CEDEFOP dataset, more than 90% of France language offers originates from 10 sources (on a total of 237 sources in the data).

This makes identifying and grouping duplicate ads an essential step in data processing. The common practice of posting the same job ad on multiple websites, unless accounted for, can result in significant over-counting of job ads.

The process of identifying and removing duplicate job ads is called deduplication. Near-duplicate jobs ads are rarely identical: when posting an ad on a different site, text is often restructured, shortened or extended and metadata is often changed to match the site's data model. Duplicated job ads can be processed by identifying a series of identically parsed variables in job ads, such as job location, employer, job title, contract type. Time frames are also important. Combining these different variables is essential in order to identify duplicates, while none of each would be sufficient by itself. For instance, consider two job ads, which only differ by their location: these ads are clearly not duplicates and should be treated as separate vacancies, unless they correspond to a unique job that is actually executed on multiple locations (Figure 8).

ID COUNTRY	DAY_GRAB_DATE	IDESCO_LEVEL_4	IDREGION	IDCONTRACT	ID EDUCATIONAL_LEVEL	ID SECTOR	ID SALARY	IDWORKING_HOURS	IDEXPERIENCE	SOURCE	SITE	COMPANY NAME	GENERAL_ID
FR	1	2433	FR24	4	5	69	NA	NA	NA	EL_SKYWALKER	wizbii	wizbii	159323362
FR	1	2433	FR53	4	5	69	NA	NA	NA	EL_SKYWALKER	wizbii	wizbii	159323362
FR	1	2433	FR61	4	5	69	NA	NA	NA	EL_SKYWALKER	wizbii	wizbii	159323362
FR	1	2433	FR10	4	5	69	NA	NA	NA	EL_SKYWALKER	wizbii	wizbii	159323362
FR	1	2433	FR22	4	5	69	NA	NA	NA	EL_SKYWALKER	wizbii	wizbii	159323362
FR	1	2433	FR82	4	5	69	NA	NA	NA	EL_SKYWALKER	wizbii	wizbii	159323362
FR	1	2433	FR43	4	5	69	NA	NA	NA	EL_SKYWALKER	wizbii	wizbii	159323362
FR	1	2433	FR71	4	5	69	NA	NA	NA	EL_SKYWALKER	wizbii	wizbii	159323362
FR	1	2433	FR62	4	5	69	NA	NA	NA	EL_SKYWALKER	wizbii	wizbii	159323362
FR	1	2433	FR26	4	5	69	NA	NA	NA	EL_SKYWALKER	wizbii	wizbii	159323362

Figure 8: Similar job adverts with different region

Furthermore, it is not possible to identify duplicate ad when missing, misclassified or unstandardized values are present in the observed variables. The main issue is therefore to properly define the selection of the data, the processing phases/s and criteria used to perform deduplication.

Request for multiple parallel processing pipeline (large amount of data, complex and time consuming data extraction, language specific data processing, the need for timely data processing) and the consequent merging of processed data can also produce duplicate job adverts. Since several data processing phases are needed deciding when to execute deduplication and how many times is important. Such procedures should be stable in time and very well documented.

OJA classification

Online job advertisements scraped from the Internet come from various job portals and enterprises' websites. Different portals often code professions with different coding systems. Therefore, it is important to use an occupation classifier to deduce for each job advertisement the standard occupation code that is necessary to calculate job vacancies by occupations.

Since last ESSnet work package some classification work on text have been carried out either for occupation or other items. Many countries considered supervised learning. One challenge was

accessing good quality data. The more categories, the more labelled data are required. For example, the international standard classification of occupations (ISCO) defines 436 occupations on the detailed level. To classify on this detailed level, a big amount of training data are needed.

Advertisements' titles and offers' descriptions (if available) are often used to classify occupation. Titles usually give a concise description of the jobs which makes them computationally less expensive to use than full description. However, sometimes titles may not contain enough information to build a satisfying classifier.

Thus, the overall challenge is to find optimal algorithms considering available training datasets, which characteristics (such as size and features) may change from one country to another. Apart from a satisfying accuracy and F1-scores, such algorithms must also be not computationally costing and should be easy to maintain.

Text classification

Text classification is well studied in some areas like information retrieval. The task is to construct a classification model (classifier), which catches the features from the training text and their relations to the class label. The model is then used to predict texts with unknown class labels. The label is either explicitly assigned with one class, called hard version; or with probability value, called soft version.

Text encoding transforms the text into a numeric vector before the modelling. First, texts are tokenized, which means divided into units where each unit is called a feature. Bag-of-words is a common language representation. By n-grams one or several words (letters) in a sequence is transformed into units. 1-gram is to use one word as a unit, 2-grams is to use two words in sequence as a unit and so on.

Feature counts, i.e. counts of the frequencies of a feature in the text, can be used as the feature weight. A piece of text is then represented with a vector of words counts in the text [22]. For example, the sentence "There are many countries participate in work package B" is transformed by 1-gram feature counts into [1, 1, 1, 1, 1, 1, 1, 1, 1], where each word appears one time in the sentence.

Tf-idf is another common method for weighting the words, which considers both the frequencies of a term in one text and the term frequencies in the whole text corpus [22].

With bag-of-words model, a big and sparse matrix is generated from corpus. Feature selection can help efficiently chose the best features and decrease the matrix size for model training. Most of the time, the feature selection is done after stop-words removal and word stemming.

Word embedding is another language model transforming text, which represents words in dense vector space and can learn words relations by consider the text sequences.

After text processing and language modelling, a classification model is chosen. Many models for text classification exists, such as decision tree, pattern (or rule) based classifiers, SVM classifiers, Neural Network classifiers and Bayesian classifiers.

Finally, the general process of text classification can be described as data gathering, data exploration, data preparation and feature selection, model training and evaluation [23]. Data gathering works on collecting and accessing to data needed for model training and evaluation. Data exploration focuses

on knowing the data, answering questions like average number of words in text corpus and distribution of classes over dataset. Data preparation focuses on text cleaning and tokenization, removing stop words, word stemming/lemmatization and feature weighting and selection, which result in numeric vectors of text. Model training and evaluation are the process of constructing models from training data and choosing the best hyper-parameters and models which are then evaluated.

Occupation classification

There are several different classifications that can be extracted from the job vacancies. The most common classification for occupations is ISCO (International Standard Classification of Occupations) or ESCO (European Standard Classification of Occupations)[5].

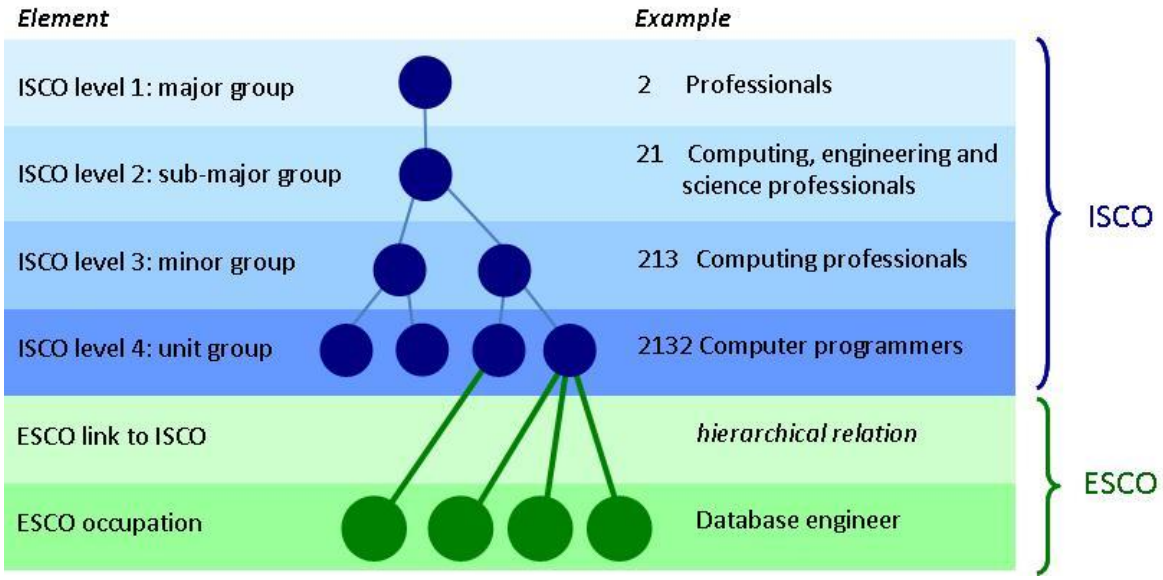


Figure 9: ESCO vs. ISCO - mapping the occupation level

ESCO classification includes 2,942 occupations (as of 5 September 2019) divided into 10 groups[6]:

- 0 Armed forces occupations
- 1 Managers
- 2 Professionals
- 3 Technicians and associate professionals
- 4 Clerical support workers
- 5 Service and sales workers
- 6 Skilled agricultural, forestry and fishery workers
- 7 Craft and related trades workers
- 8 Plant and machine operators and assemblers
- 9 Elementary occupations

[5] https://ec.europa.eu/esco/portal/escopedia/ESCO_in_statistics, as of 5 September 2019
[6] https://ec.europa.eu/esco/portal/occupation, as of 5 September 2019

Each occupation contains one preferred term and any number of non-preferred terms and hidden terms in each of the ESCO languages. Each occupation has its own profile, which contains an explanation of the occupation in the form of description, scope note and definition. They also list knowledge, skills and competences needed for each occupation. In full view of the ESCO classification it includes:[7]

- Description
- Alternative label
- Regulatory aspect
- ISCO-08 code
- Definition
- Scope notes
- Hierarchy
- Narrower occupations
- Essential skills and competences
- Essential knowledge
- Optional skills and competences
- Optional knowledge
- Status
- Replaces
- Replaced by
- Concept URI

This wide scope of the definition of each occupation does allow for a more refined narrower classification.A snapshot of the 4-level hierarchy according to the CEDEFOP dataset is presented in Figure 10.

[7] https://ec.europa.eu/esco/portal/occupation, as of 5 September 2019

	idesco_level_4	esco_level_4	idesco_level_3	esco_level_3	idesco_level_2	esco_level_2	idesco_level_1	esco_level_1
0	2634	Psychologists	263	Social and religious professionals	26	Legal, social and cultural professionals	2	Professionals
1	2352	Special needs teachers	235	Other teaching professionals	23	Teaching professionals	2	Professionals
2	4321	Stock clerks	432	Material-recording and transport clerks	43	Numerical and material recording clerks	4	Clerical support workers
3	7131	Painters and related workers	713	Painters, building structure cleaners and rela...	71	Building and related trades workers, excluding...	7	Craft and related trades workers
4	2512	Software developers	251	Software and applications developers and analysts	25	Information and communications technology prof...	2	Professionals

Figure 10: ESCO different levels representation in CEDEFOP dataset

There are two common methods to map the occupation with the classification code:

1) Text mining
2) Machine learning

Both methods have some common tasks to do before the matching algorithm is used. It is presented in Figure 11.

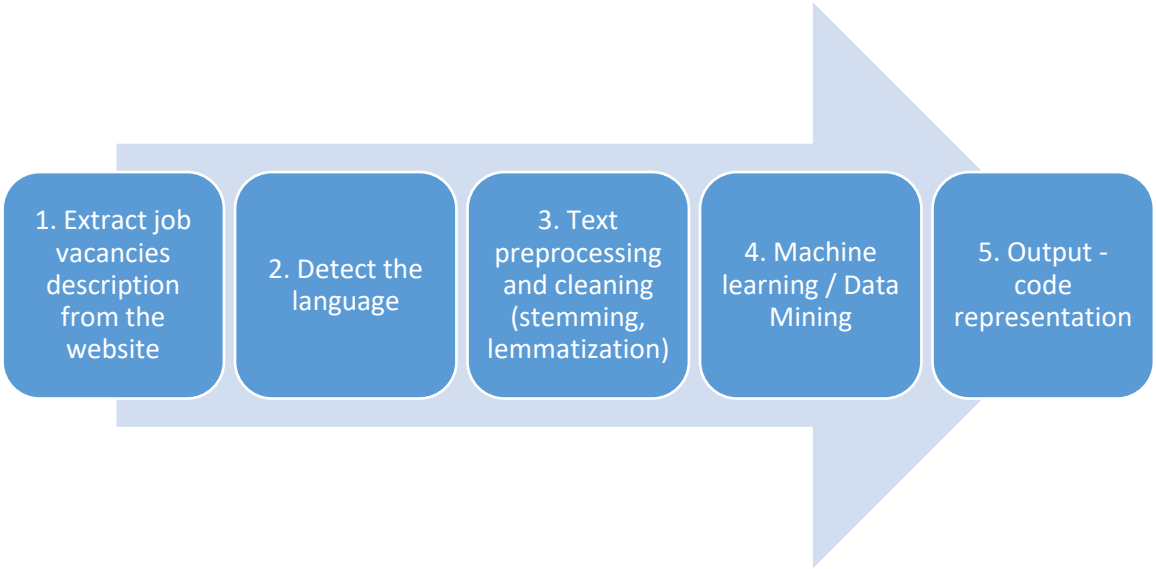


Figure 11: General steps in the process of ESCO code detection

To start the process of classifying an occupation, it is necessary to extract relevant data from the website. Usually it is a short job description. The second step is to detect the language. It is important to match the language with the job offer. Otherwise, the occupation code will not match correctly. The third step is to clean the data – it includes stop words removal, stemming and lemmatization, briefly described in the previous section of this methodological framework.

The fourth step is to use the machine learning or text mining algorithms to match the description of the job offer to the proper ESCO code. In the first case, a training dataset has to be created and defined

for each language separately. The last step is to provide the code representation of the occupation for the detected job vacancy, in hierarchical form, as presented in Figure 11.

According to the architectural business functions used in this methodological framework, the following business functions are involved in the occupation classification process:

- Acquisition and Recording
- Data Wrangling
- Data Representation

Training occupation classifier

Two NSIs trained occupation classifiers: Dares (France) and Statistics Sweden. Quality levels obtained by France and Sweden differ a lot: accuracy reaches 0.92 in French case and 0.76 for Swedish experience. This might be due to different data set sizes, as well as differences in data modelling and machine learning algorithms (SVM vs. Neural network). Both experimentations are described below.

DARES

Since September 2018, Dares collects online job offers from multiple websites on a daily basis. Dares needs to classify job offers it collects from the Internet into the French occupation nomenclature, abbreviated ROME (from the French version Répertoire Opérationnel des Métiers et des Emplois), in order to produce statistics such as labour market tightness, broken down by occupation. Thus, a SVM classifier was trained and tested using more than 1,500,000 labeled offers published on the French National Employment Agency (in French, Pôle emploi) in 2019. On test data, F1-score and accuracy of this 532 labels' classifier reaches 0.92. Although many improvements are to be made, Dares considers those results good enough to publish statistical outputs such as labour market tightness indicators using automatically labelled offers.

Table 6: ROME occupation nomenclature insight

Occupation level	Number of classes	Code example
Level 1 *(Grands domaines)*	14	**H** (Industry)
Level 2 *(Domaines professionnels)*	110	**H12** (Industrial engineering studies and R&D)
Level 3 *(Code ROME)*	532	**H1204** (Industrial design)

Data gathering

The dataset contains a collection of ~1,547,800 job offers published between January 23rd 2019 and December 31st 2019 by employers on the French National Employment Agency website. Although collected job offers contains many variables, so far we have used job offers' titles and occupation codes (code ROME) only to train our model.

For each occupation code, French NEA provides a list of job titles (total 11,081). Since all occupations were not covered by our dataset, provided examples were added, which led to a dataset of ~1,558,800 examples.

The dataset is unbalanced: gathered data distribution largely differs from ROME occupation codes' distribution within the nomenclature (Figure 12).

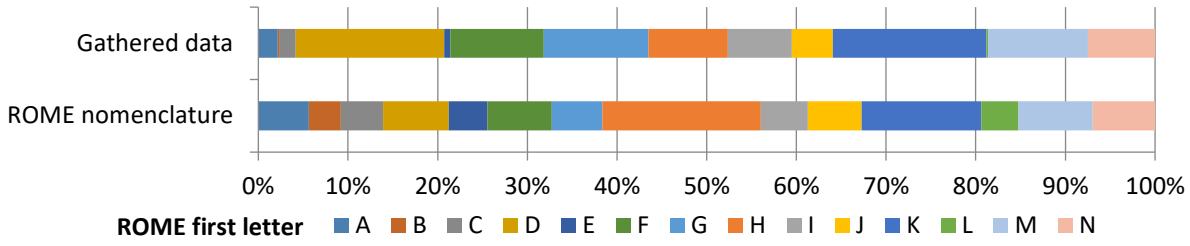


Figure 12: ROME level 1 distribution in dataset vs nomenclature

Among each level 1 class, occupation codes are also not equally represented (Figure 13). The mean number of examples per code is 2930, the most represented occupation code has 43,767 examples and the less represented one 7.

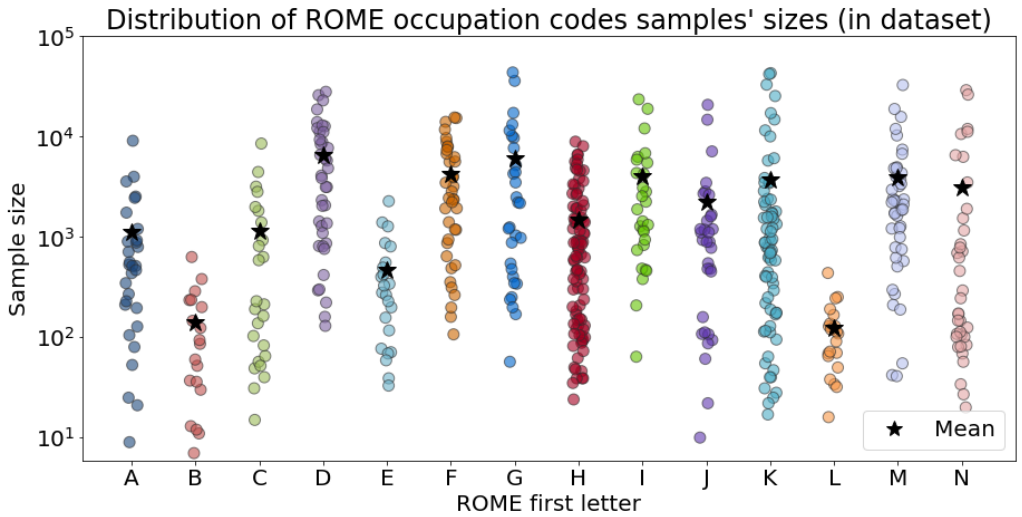


Figure 13: ROME occupation codes samples' size distribution

The data set is not re-balanced, since scraped job offers advertisements (i.e. what is to be classified) are likely to be imbalanced in a not so different way form training data – even though we don't claim our dataset to be representative.

Text preparation

Even if data appears to be of good quality (employers seems to check quite carefully job offers' information before publishing them), some text processing is required to normalize titles and reduce vocabulary size.

Job offer titles go through the following process:

- Texts are standardized (which includes conversion to lowercases, removal of punctuation and numeric characters)
- Stop-words and parasite job related words (such as contract type or working hours) are removed. Location information such as country, region or district names are also removed. (City names are not removed).
- Titles are lemmatised with Morphalou lemmatizer [19] (only available for French language). We also lemmatize female job nouns to male equivalent (eg. "actress" becomes "actor"), since French language tends to decline all job nouns by gender. Finally, we expand abbreviations

and acronyms, using ~150 job related acronyms or abbreviations extracted from gathered data and NEA sources.

- For each title we keep only one occurrence of each word, since repetitions are mostly due to lemmatisation (eg. “Actress/ Actor” becomes “actor actor” which is transformed into “actor”).

Vocabulary insights

The vocabulary size decreases at each processing step. From 31,991 unique tokens in raw data, vocabulary is narrowed to 22,019 words after text processing. After processing, empty titles are removed from the dataset and the mean number of tokens per title is 2.7.

More than half words in the vocabulary (54.9%) occur in a unique occupation code in our dataset. Apart from code ROME specificity, it can also be due to misspelled words that are used once in the whole dataset and which are not dealt with for now.

Selected model and results

After testing a Text Similarity algorithm which was described in the first ESSNet report [2] and which was not very efficient, we looked at Machine Learning algorithms. As SVM algorithms were providing the best results on a similar problem of classifying job offers from their full text [20], we chose to focus on this algorithm.

Data is split into a train and a test set (80%/20%) and titles are transformed into bag of words vectors. Then, a linear SVM classifier is trained (with hyper-parameter C = 1) in Python, using the well-known Scikit-learn library [21]. This multi-label classifier is trained to classify 532 codes, but we compute scores at different occupation levels (Table 7). Computed scores are accuracy, which is the rate of correctly labelled samples, and F1-score defined as the harmonic mean between precision and recall for each code. We then compute weighted and macro F1-score as the weighted and unweighted average of F1-scores for each code, respectively. At the less precise level (level 1) accuracy and weighted F1-score reach 0.96, which of the most satisfying considering that unweighted F1-score is 0.94. On ROME codes, F1-score is 0.92, but the unweighted F1-score tends to significantly drop (-0.10 from level 1 to level 3).

Table 7: SVM linear scores on test set

Occupation level	**Accuracy**	**F1-score** *Weighted average*	**F1-score** *Macro average*
Level 1	0,96	0,96	0,94
Level 2	0,94	0,94	0,87
Level 3	0,92	0,92	0,84

When focusing on each code (Figure 14), it appears that the classifier is less performant on classes lacking samples per code (eg. B, E and L, for which dataset contains less than 500 samples per ROME code on average (Figure 8)).

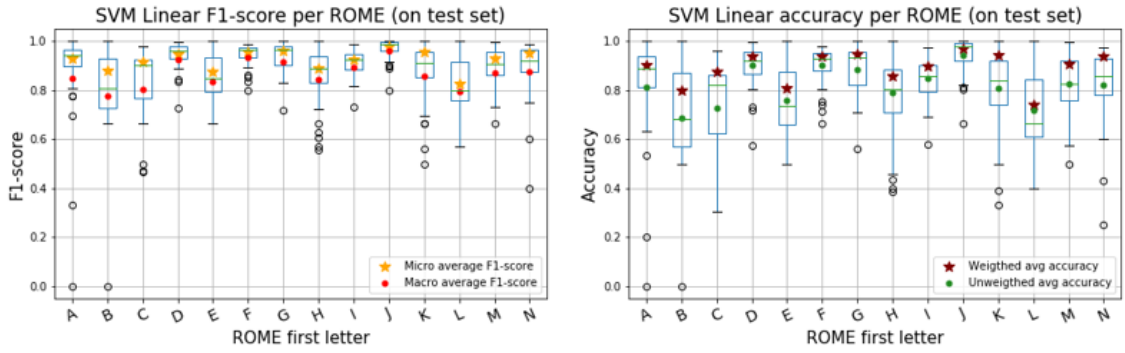


Figure 14: F1-score boxplots (left picture) and Accuracy boxplots (right picture)

Confusion matrix (Figure 15) shows that classes with more examples per occupation code are more wrongly predicted. 41% of misclassified titles (at level 1) are labelled as D, G, F or I (top four mean samples' sizes classes in dataset) when they contain 23% of nomenclature's occupation codes.

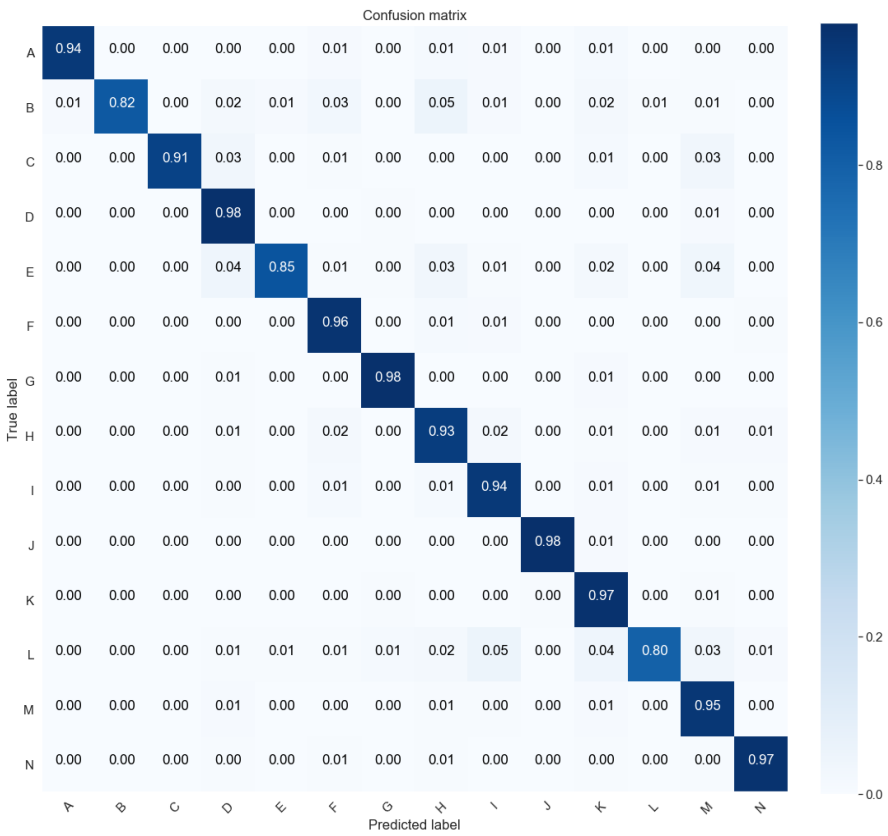


Figure 15: Normalized confusion matrix

Future work

Apart from testing other ML algorithms, we are confident that our classifier can be improved. As word order matters in NLP, and its importance may be enhanced in short texts, the first improvement would be to take into account the sequence of words instead of using a flatten bag-of-words representation.

A second improvement could be to take advantage of the hierarchical structure of our nomenclature and train multiple models to classify data "level by level". The idea is to train one model to classify the

first letter of the occupation code. Then 14 models (one for each letter) would determine the two next digits and finally, last models would predict the full occupation code. The main benefit of such a classification pipeline would be to reduce the number of classes to predict at each step, instead of classifying 500+ classes at once.

Finally, although it is computationally more expensive, using offers' full description when classification from title is not possible or ambiguous is being studied. It would lead to a more rule-based classification and requires a detailed analysis of results at the finest level (level 3) to precisely understand which categories or features are problematic.

Sweden Statistics

Sweden Statistics classify 46 occupations, i.e. two-digit-level of the standard Swedish professions' code (SSYK), which is developed from the international standard ISCO by Statistics Sweden. Data come from the advertisements of the Swedish Employment Service, which contains the variable 'title of advertisement' and 'SSYK code'.

Following the instruction of text classification from Google [23], one-gram model is applied and a linear stack of layers neural network model is constructed. Word embedding has also been tested.

Through choosing different sample sizes and compare the results, we come to an optimal sample size considering the training time and the accuracy requirement. Feature selection is very important for efficient training.

The overall accuracy comes up to 72% with 2000 samples per class, which is an improvement comparing with our previous tests.

Data gathering

The data are gathered from Swedish Employment Service (SES), containing the variable "title of advertisement" and "SSYK" assigned by SES. In total, there are more than 4 million advertisements from 2007 to now. The distribution of the 46 classes are not even in the dataset, occupations such as requiring university education in the area of education, health and medicine and occupations for sales are many; some occupations have less than 1000 sample, such as manager in financial and bank institutes and special officer.

Two datasets of 1000 and 2000 samples for each class are drawn randomly from the total advertisements by applying the bootstrapping with replacement procedure, see Table 8.

Data exploration

In average, the number of words in each advertisement's title are three to 4.6 words. The length of words in the titles are between one to more than 20 words. Some titles are ambiguous, not containing enough information for classification. This can been seen in words distribution of the clean text corpus, some titles become empty strings after stop words removing and word stemming.

Table 8 shows the two datasets in the test.

Table 8: Training data set description

Text	2000 samples	1000 samples
Number of samples	92 000	46 000
Number of classes	46	46
Number of samples per class	2 000	1 000
Average words per sample	4.6	3

A very small sample has been examined manually by the domain expert to see the matching quality of the occupation label. The expert found examples with not enough and ambiguous information in the sample. Otherwise, the SSYK coding in the dataset is good.

Data preparation

The text corpus is cleaned, i.e., first set characters to lowercase, tokenizing text into unit delete illegal characters and numbers from the text, and then delete stop words that do not contribute to the occupation classification; and last stemming the words.

After the cleaning procedure, 16 titles in 1000-sample and 38 in 2000-sample dataset become empty strings. The average number of words in both datasets are 2.8, and the variation is between zero words to maximum 10 and 11 words. The number of words' distributions in the samples are presented in Figure 16; about 90% of samples contain one to five words in both 1000 samples and 2000 samples.

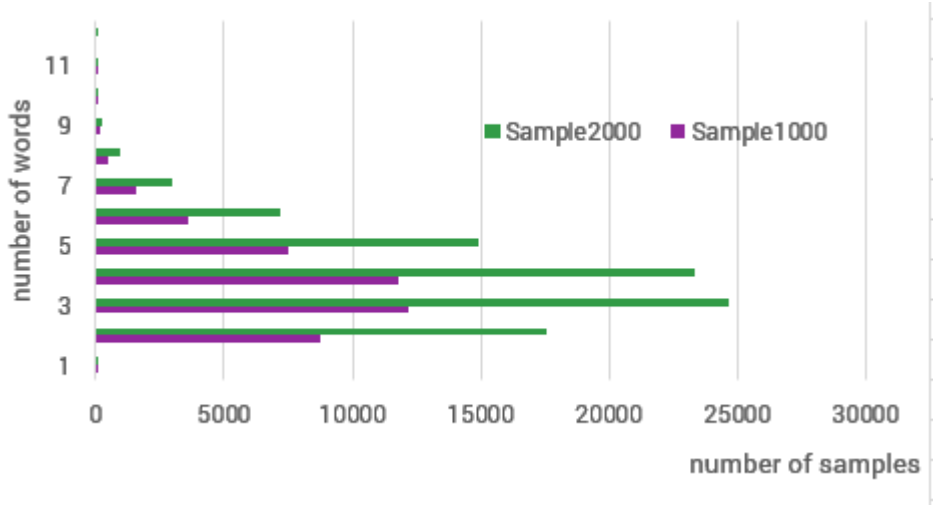


Figure 16: Text length distribution

The visualization of the most 100 common words in 1000-sample dataset is shown in Figure 17 below. We choose 1-gram word and most of the words are occupations. The words are mainly occupations that do carry the occupations meaning.

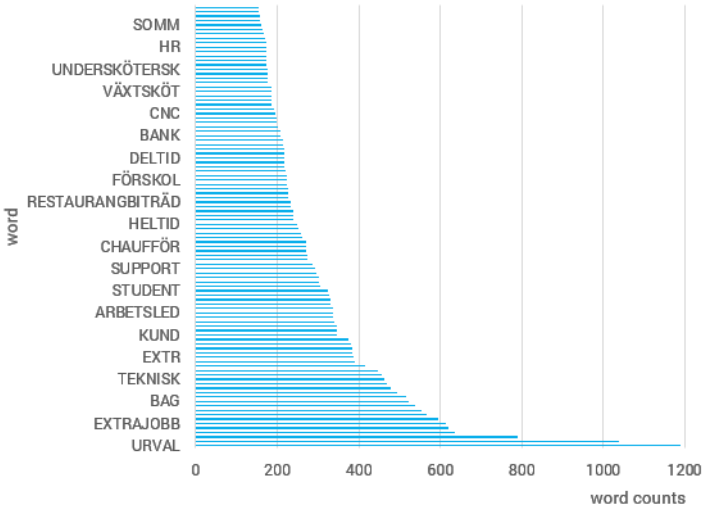


Figure 17: Most important words

Feature selection

Feature selection is an automatic way to select the best features for model prediction. According to the rule recommended by Google, ratio of "number of samples" and "number of words per sample" when the value is under 1500, n-gram model is preferable considering the accuracy and the computation time. This is our cases, therefore 1-gram model is applied and tf-idf is calculated for representing words. Chi2 is applied to choose the best features from the training text corpus and labels. We examine the scores of the most influencing words in the model; many occupation wordss are included in the most import features, which can be interpreted that Chi2 feature selection is very effective. By feature selection, not only the important words are chosen, but the text vectors are in much smaller dimension, 4000 features instead of ~11000 features.

Build model and tuning the hyper-parameters

Both datasets are divided into training and evaluation datasets by 67% vs 33%. The neural network model consist of a linear stack of layers. The dimension of the input layer comes from the input data shape, the dimension of the output layer is how many classes there are in the dataset. The intermediate layers are also called hidden layers, of which dimensions of hidden layers are handled automatically. Other parameters specific for multiple classification problem are the activation function of the last layer should be "softmax" and the loss function in the model should be "categorical_crossentropy".

Figure 18 and 19 show the searching of the best hyper-parameters of layers and features. The two figures show similar structure, layers of two return high accuracy result, the features can be limited to 2500 to 4000. Figure 14 shows that feature 10000 does not improve much comparing with feature 4000. The results show that it is worth to set layers to two and number of features to 4000.

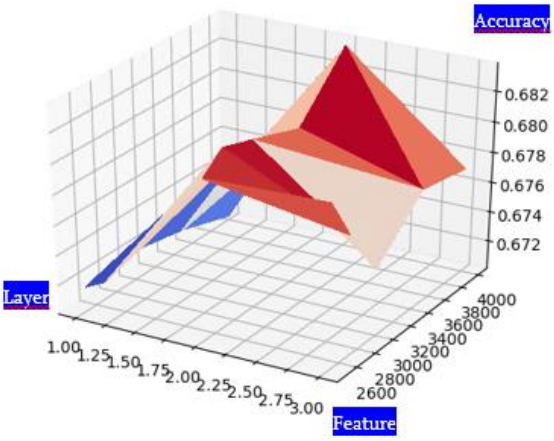


Figure 18: Tunning hyper-parameters, 1000 samples per class

Figure 20 shows the accuracy training history of the two datasets with the model containing two layers and 4000 features. The convergence is very fast, after 1 or 2 epochs. By setting epochs to around 10 saves training time. When samples size is increased into 2000, the accuracy does improve from 65% to 72%. However, the models have slight overfitting problem, i.e., the training accuracy is higher than the test accuracy.

The accuracy increases not much by epochs, we need to add more words, such as words describing skills and competence. Again, it is a challenge even to access such dataset, that the skills and competence are labelled to occupations' label.

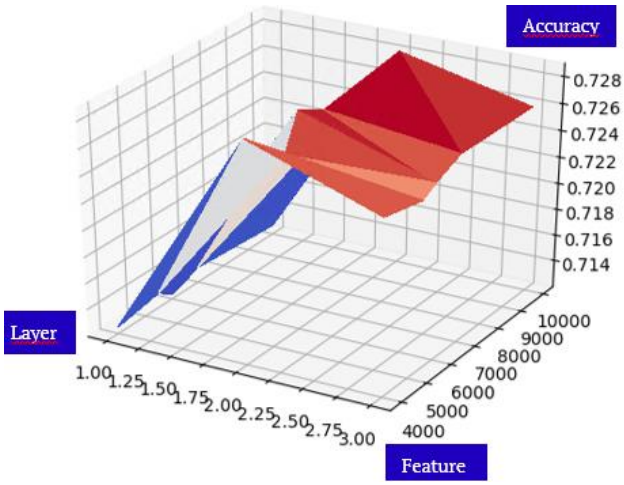


Figure 19:Tuning hyper-parameters, 2000 samples per class

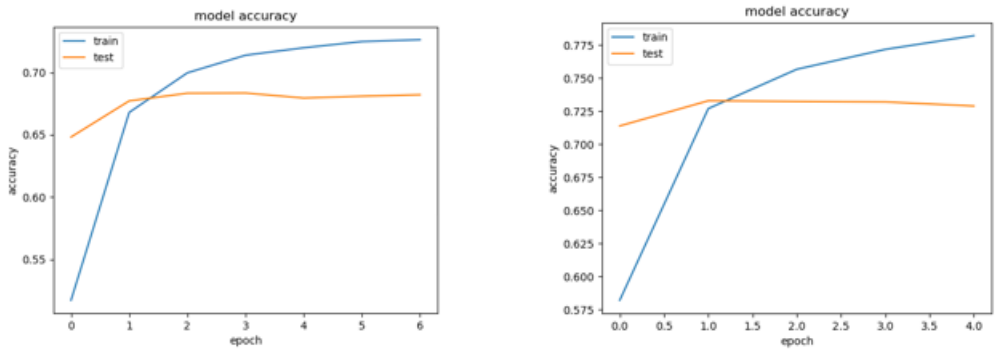


Figure 20: Model accuracy in learning process (left: 1000 samples per class; right: 2000 samples per class)

The tests on the two datasets show that the accuracy still has a space for improvement. The samples were increased for each class to 5000. According to the ratio between "the number of samples" versus "the number of words per sample", the model option for the 5000 samples is word embedding. With the word embedding as input, we tested the same 2-layer linear stack neutral network model. 100 dimension were chosen as the word embedding, and added a tensorflow embedding layer in the model. The accuracy learning history is shown in figure 21. Through the epochs, the accuracy has been increased, the overall accuracy has been increased to 76%.

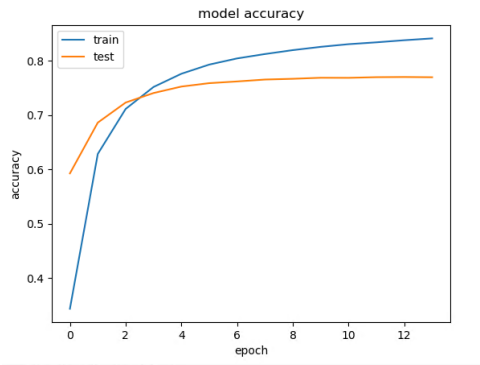


Figure 21: Model accurcy in learing process with word embedding, 5000 samples per class

Results and future work

Of the 46 classes, the prediction accuracies are all above 50% except for one class. The correct predictions of every class have the highest scores, shown in figure 22. The overall accuracy of the model has been improved significantly comparing with our previous tests, up to 76% accuracy. Adding chi2 feature selection in the data preparation process, the training becomes more efficient for the same result. It seems that chi2 feature selection gives the similar effects as the word embedding.

The accuracy has also been improved by increasing the sample size, 1000 samples reaches 65%, 2000 samples reach 72% and 5000 samples come up to 76% accuracy. By considering the trade-off of accuracy, overfitting and efficiency of training, 2000 samples and the 1-gram model should be the best choice. From the data preparation, feature selection, model building and hyper-parameter tuning, the procedure return the accuracy results acceptable.

The classification model needs to predict on the detailed occupation level, for which we need to try on even larger sample size.

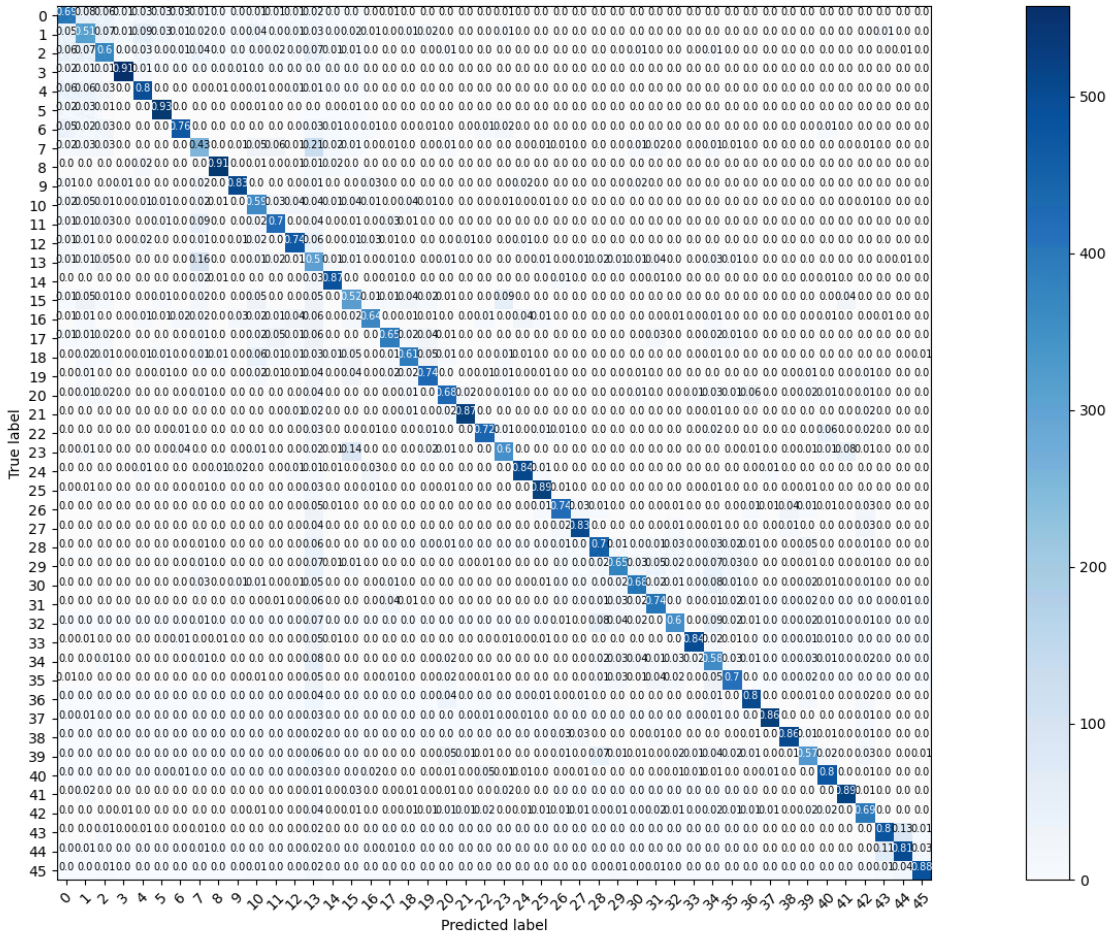


Figure 22: Confusion matrix of 46 classes, 2000 samples

Another issue is the metric of accuracy used for model training. There are recall and precision metrics available that we should consider for a better model construction.

Methodological description on fitting discrete probability distributions combining jobs survey data with OJAs data

In contemporary society, based on observations made so far, we can make the following axiomatic assumption: data comes in different ”shapes and sizes”. Here by data we denote one or more random variables. If this is the case, a fundamental problem in statistics is derived from the natural question: “What data generation process, i.e. mathematical function, optimally approximates our data?”. This is a very hard question, either if we look at the type of the problem from the perspective of a search problem on an infinite space of solutions, more so if we look into at the mathematical theory which makes this type of problem tractable. In general, there is no silver bullet to this type of problems, and a variety of heterogeneous strategies are used from many fields of mathematics, e.g. ranging from computing some measure on the sample space to solving some partial differential equation. A secondary problem which can be derived from finding the functions, is given by the question: “if two random variables are approximated by distributions within the same family, can we approximate the random variable for which we don’t have observations at a certain time instant, with another variable for which we have observations?”. This last question is of interest in the case of estimating the unobserved number of job vacancies in terms of the observed number of online jobs advertisements.

Although it is not necessary to grasp the mathematical reasoning behind the notions in order to apply the algorithms to generate feasible solutions, it is generally recommended to have at least some basic notions of probabilities and mathematical distributions. An excellent primer in theoretical probabilities and statistics is the Cassela & Berger textbook [24] in which most of the mathematical statistical concepts employed in this work are thoroughly explained, with rich examples.

One important tool in the belt of a practitioner statistician is the procedure of fitting a known family of theoretical probability distribution to a dataset, i.e. one or more random variables, generated by a process which entails, or not, independence between observations and/or variables. This is mainly a three-step iterative algorithm and consists in making an informed/educated guess about the theoretical family which best fits the data, estimating the parameters for the theoretical distribution and testing the null hypothesis which states that the parameters of the theoretical distribution optimally approximate the parameters of the empirical distribution, i.e. the differences between the two types of parameters asymptotically converge to 0. If the theoretical probability distribution optimally fits the empirical one, then different types of statistics about the dataset can be estimated and/or used in mathematical models which assume a certain type of probability distribution. This whole procedure is, more or less, abstracted/simplified in figure 23.

The first step in applying the distribution fitting procedures to a dataset is to make some sort of educated guess regarding what family of distribution, i.e. analytical form, best describes the generation process of the dataset. Here, two observations are in order to understand the difficulty of this process. Firstly, finding a good approximation for the empirical distribution is very hard, in large part due to the intractable number of probability distributions. Secondly, the problem is domain specific, requiring domain knowledge input. The second step involves estimating the parameters of the theoretical distribution based on the observations. At this step, a procedure called maximum likelihood estimation is used. For the third step some sort of measure for the goodness-of-fit is applied to test if the differences between the parameters for the empirical probability distribution function and the estimated parameters of the theoretical probability distribution function are statistically different from 0. If this is the case, then population parameters can be estimated, otherwise the procedure jumps to step 1, in which a different selection is made.

In the following sections, each of the steps will be presented along with examples using datasets collected as result of the project (national collected datasets) and already published national statistics on job vacancies. Also, we will provide a custom R function which will try to fit some theoretical distributions to a dataset comprising two categorical variables (count data), the official statistics regarding the number of job vacancies broken down by some type of classification rule, and an accurately classified according to the same rule. As an output this function will generate a pdf (portable document format) report.

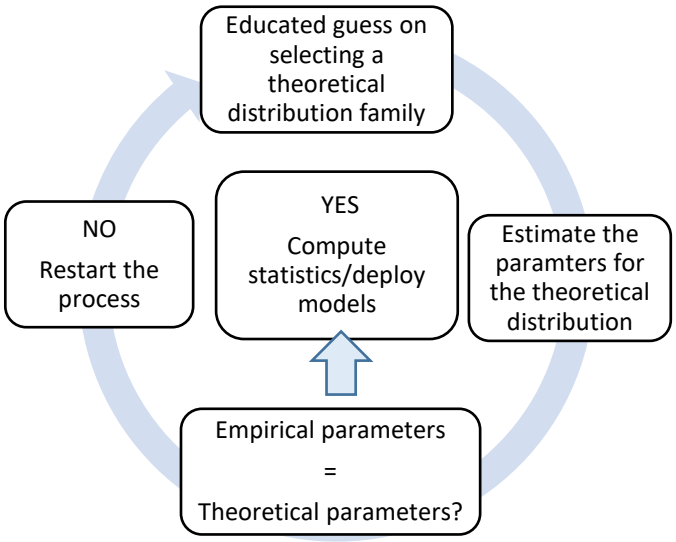


Figure 23: Simplified overview on how to fit a distribution to a dataset

Theoretical background for the estimation procedure.

The procedure used to fit a dataset to a probability distribution function, through estimation of the parameters is known as maximum likelihood estimation.

Maximum likelihood estimation (MLE) (Cassela & Berger, 2002) [24] is a statistical technique for estimating the parameters of probability density distribution based on the likelihood function, which in a frequentist interpretation postulates that all the relevant information in a sample is contained in the likelihood function, i.e. "data are interpreted as evidence, and the strength of the evidence is measured by the likelihood function".

Main steps in MLE:

1. Make an educated guess about the probability distribution function (p.d.f.) which optimally fits the data.
2. Compute the likelihood function given by the formula:

$$\boldsymbol{L_n(\theta) = \prod_{i=1}^{n} f(x|\theta)}\,,$$

   Where

$$\boldsymbol{L_n(\theta; x)} - likelihood\ function,$$

$$f(x|\theta)\text{- pdf}$$

$$\mathbf{x} - \text{random variable or sample data,}$$

$$\boldsymbol{\theta} - paramter(s)\ of\ the\ p.d.f.$$

   Given the nature of this function in practice is simpler to compute the log-likelihood function, therefore the former can be expressed

$$\boldsymbol{ll(\theta; x) = \sum_{i=1}^{n} \log(f(x|\theta))},$$

   Where

$$\boldsymbol{ll(\theta)} - log - likelihood\ function$$

3. Estimate the parameter(s) θ which maximize $ll(\theta)$, formally

$$\boldsymbol{\widehat{\theta} = \arg max\ ll(\theta; x)}$$

This is usually achieved by solving the Langrage multipliers system of equations.

Dataset description

The input dataset consists of three variables, namely Romanian NACE Section codes (main types of industries) [25] consisting in uppercase letters, denoted by NACE, the number of job vacancies from national survey by NACE Section code [26], denoted by JVS, and the number of jobs advertisements collected online from the Romanian National Agency for Job Vacancies [27], denoted by OJA. The JVS consists of quarterly aggregated data by NACE codes (sections, divisions, etc.), for the first quarter of 2020, which was aggregated in a set of 19 observations by NACE Section codes. The OJA consist in 17 observations (given the period of the year, there is missing – NA - data on some NACE Section codes, e.g. agriculture) from data collected on 28 March 2020, also aggregated on NACE Section codes. The OJA data is a tidy dataset by design, so this estimation exercise assumes that the dataset is optimally classified, already classified by NACE codes. The two datasets are joined in a common structure by NACE Section codes, resulting in 19 observations. Some descriptive statistics for our dataset are presented in table 9. Descriptive statistics for input dataset.

Table 9: Descriptive statistics for input dataset

Descriptive statistics	JVS	OJA
Minimum	56	66
1st quartile	397	160
Median	1008	446
Mean	2198	1126
3rd quartile	2910	1355
Maximum	9230	6267
Missing	-	2
Standard deviation	2498.039	1677.894
Inter-quartile range	2513	1195
Coefficient of variation	1.1365	1.4895

Estimation and software tools

For testing the optimal fit of a statistical distribution, we used fitdistrplus R package [28], which provides an easy to use interface for testing the goodness-of-fit for several popular probability distributions families. The easiest way of selecting a probability distribution is through descdist function, which compares the squared skewness, the third statistical moment, and kurtosis, the fourth statistical moment, of the empirical distribution with those of theoretical distributions, for continuous or discrete probability distributions, and computes parameters of the empirical distribution, minimum, maximum, median, sample mean and standard deviation. From the output of this function a Cullen-Frey [29] graph is produced in which is easily identifiable membership of the empirical

probability distribution to one of the classical families. In figures 24 and 25 Cullen-Frey plots are presented for job vacancies from the national survey and for job vacancies advertisements collected from the Romanian National Agency for Job Vacancies.

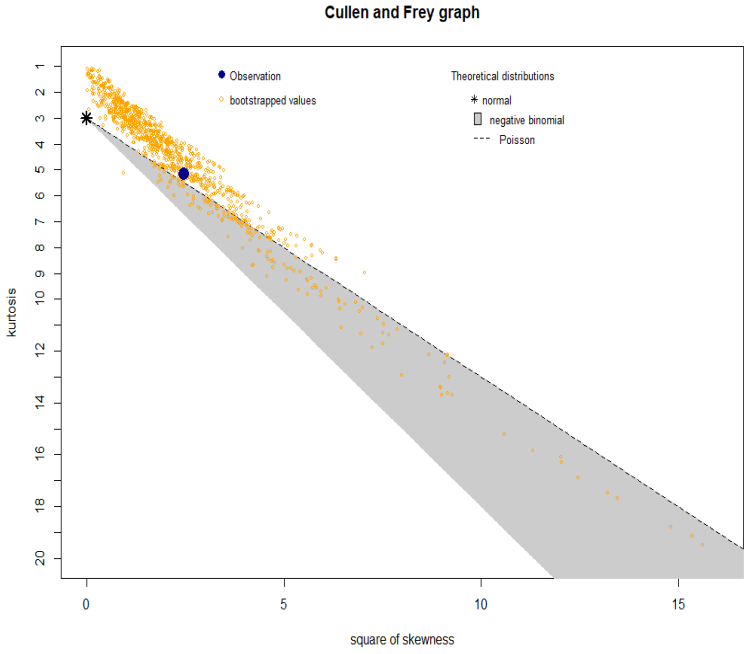


Figure 24: Cullen- Frey graph for job vacancies. Dataset: Romanian NIS Job Vacancies Survey.

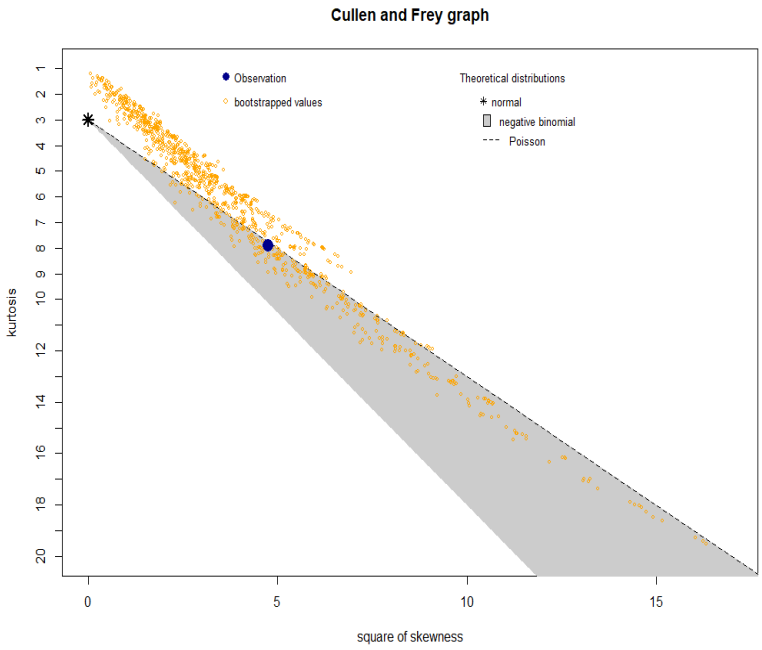


Figure 25: Cullen-Frey graph for job advertisements. Dataset: Romanian National Agency for Job Vacancies.

The two graphs describe how "close" the empirical probability distributions are to a few discrete probability distributions. The blue point represents the coordinates of the squared skewness and kurtosis computed by 1000 bootstrapped samples.

Based on the two graphs, we can try to fit a negative binomial distribution to our dataset. The negative binomial distribution has the following probability mass function:

$$\boldsymbol{f(x,k,r) = \binom{k+r-1}{k} p^r (1-p)^k}$$

where,

$$\boldsymbol{k} - non - negative\ integer\ support; number\ of\ failures$$

$$\boldsymbol{r} - non - negative\ \ integer;$$

$$number\ of\ succeses\ or\ the\ dispersion\ parameter\ of\ the\ gamma\ mixing\ distribution$$

for which we can compute skewness (third-order moment):

$$\boldsymbol{\tilde{\mu}_3 = \frac{1+p}{\sqrt{pr}}}$$

where,

$$\boldsymbol{p} - is\ the\ probability\ to\ record\ r\ succeses$$

and the kurtosis (fourth-order moment)

$$\boldsymbol{\tilde{\mu}_4 = \frac{6}{r} + \frac{(1-p)^2}{pr}}$$

upon which the probability distribution family was selected. The negative binomial distribution can be regarded a Poisson mixture with the rate parameter, λ, generated by a gamma probability distribution. In this case, the parameter of interest is r, number of successes or dispersion parameter. This solves the problem of constant rate λ and under/over dispersion can be modelled with ease by a negative binomial probability distribution.

The next step involves estimating the parameters through MLE. The parameters for the negative binomial distribution are the number of trials until the *r* success and the probability *p* for success.

In this case, the estimated parameters are presented in table 10. Estimated parameters for theoretical probability distribution fitting, where we can observe a greater variability of the JVS compared to OJA, which can entail an under coverage present in the data coming from the Romanian National Agency for Job Vacancies

Table 10: Estimated parameters for theoretical probability distribution fitting.

Variable	Parameter	Estimate	Standard Error
JVS	Size = r	0.8015	0.2246
	Mean	2198	563
OJA	Size = r	0.6415	0.1862
	Mean	1127	341

Using the estimates output we are now able to generate probabilities for each observation in our two variables, JVS and OJA.

Conclusions and further research plans

Estimating the probability density functions, under the simplest assumptions, i.e. data is generated by a univariate process, is a rather difficult problem. There are a lot of probability distribution functions, with different types of parametrisation and/or interpretations, given a problem from a certain domain. The standard procedures for fitting a probability distribution function to a dataset involves estimating the parameters of the theoretical probability distribution which optimally approximates our dataset and compare them to the parameters of the empirical probability distribution. If the differences between the two converge to 0, we can approximate our dataset with a known probability distribution. The main mathematical procedure used to compute the parameters is by maximum likelihood estimation.

The two data sources deal with count data. For count data, we have a lot of probability distribution from which we can select an optimal one. Thankfully, this process is quite easy using the R package fitdistrplus. By comparing the squared skewness and kurtosis of our empirical probability distribution with the same moments computed for a few classical probability distributions, an educated guess can be made.

Acknowledging that we barely scratched the surface on this type of inference problems, we are optimistic that new statistics can be developed by incorporating new data sources through statistical inference procedures. If this will be the case, then it can be quite easily to generate multiple types of statistical measures of uncertainty in the estimation procedures. One possible future research problem would be to estimate the joint probability distribution of the two variables and compute the conditional expectation: given a number of observations from our OJA for a certain NACE section, what is the expected value of JVS for that NACE section, formally stated as:

$$\boldsymbol{E}[\boldsymbol{Y_i}|\boldsymbol{X_i} = \boldsymbol{x_i}] = \sum \boldsymbol{y_i} * \boldsymbol{p}([\boldsymbol{Y_i}|\boldsymbol{X_i} = \boldsymbol{x_i}])$$

where,

$$\boldsymbol{Y_i} - random\ variable\ JVS\ for\ a\ certain\ i - th\ NACE\ section$$

$$\boldsymbol{X_i} - random\ variable\ OJA\ for\ a\ certain\ i - th\ NACE\ section$$

$$\boldsymbol{p}([\boldsymbol{Y_i}|\boldsymbol{X_i} = \boldsymbol{x_i}]) - conditional\ probability\ of\ observing$$

$$a\ certain\ number\ of\ OJA\ in\ i - th\ category,$$

$$given\ we\ detect\ a\ number\ of\ OJA\ in\ the\ same\ NACE\ category.$$

To compute the conditional probability, we simply use the conditional probability formula:

$$\boldsymbol{p}([\boldsymbol{Y_i}|\boldsymbol{X_i}]) = \frac{\boldsymbol{p}(\boldsymbol{Y_i} \cap \boldsymbol{X_i})}{\boldsymbol{p}(\boldsymbol{X_i})}$$

From classical Job Vacancy surveys to OJA censuses: a tentative sketch

Integrating OJA in the production of JVS and replacing JVS by OJA may be major goals of collecting OJA in future years. Such an approach would significantly alleviate the survey burden on enterprises and would also have important consequences on survey designs, with classical sampling surveys being replaced by censuses. In this section we try to:

a) propose tentative sketches of integrating OJA in the process of JVS production
b) identify necessary requirements needed to assure standards of quality
c) identify pitfalls to avoid and issues to address

Possible levels of integration of OJA in the process of JVS production

Three levels of integration are considered here:

a) Use OJA as a complementary data source to help filling non-response items
b) Replace the classical data collection using OJA in a classical sampling approach
c) Replace both the survey data collection and the sampling approach by a census approach

Corresponding sketches

Collect OJA as an additional data source to help filling non-response items in a classical JVS approach

Collecting OJA from the web as a complementary data source in order to help filling non-response items in a classical JVS approach is a goal which is easy to achieve and easy to implement in practice. Effects of non-response items can be corrected in two ways, i.e. by imputations or adjustment of survey weights. However, imputation of missing values for job vacancies is not an easy task because auxiliary variables for job vacancies are usually no available and because of the peculiar frequency distribution of job vacancies among enterprises. In addition, adjustment of sampling weights does not apply to very large enterprises which are often so dominant in their strata that they may be exhaustively sampled (treated as a census) in many surveys. Using alternative data such as OJA is therefore particularly valuable in these circumstances.

Quality criteria:

Achieving excellent levels of quality is a key issue while mixing different data sources. One should evaluate on common grounds how data from two sources may diverge from each other. Depending on the circumstances, correction factors for over- or under-estimation could be computed from paired observations from the two data sources. Evaluating the range of the discrepancies between JVS survey data and OJA, as well as understanding the causes of the differences is essential in such an approach. The more likely causes of differences are errors in reporting job vacancies, differences in the perimeters of the reporting units (especially likely in large enterprises or groups of enterprises controlling many establishments) and matching errors.

Replace the survey data collection using OJA as a substitution data source for the sampling units

Such an approach could be used in order to alleviate the survey burden on enterprises, a valuable goal when alternative data are available. A peculiar care should be put in the matching, otherwise estimates could largely diverge from those of a classical survey. A transition period using simultaneously both approaches should be planned in order to secure the quality of the results produced as well as the continuity of the time series.

Replace the sampling approach by an OJA census

This approach is similar to the precedent, except for the fact that OJA from all enterprises should be crawled. Here again correct matching is a critical issue in order to properly establish the boundaries of the relevant information. The following sketch is suggested:

a. Define the data that should be acquired, their metadata and the quality standards to achieve
b. Develop a crawling project or secure a sustainable contract of data acquisition
c. Plan a transition period to assess and reach the required quality standards of an OJA approach
d. Conduct simultaneously both approaches (classical survey and OJA) during the transition period
e. Match enterprises with units of the register (enterprises and local units)
f. Evaluate the quality of matching by simultaneously recording JV from the enterprises during the transition period
g. Dispatch OJA among local units (establishments)
h. Produce and assess estimates at all levels of aggregation between the two approaches during the transition period
i. Move to the OJA census once all quality criteria are satisfied
j. Carefully secure the transition by assessing the possible gaps between the two approaches in order to assure the continuity of time series

Pitfalls to avoid and issues to address

Secure data matching quality

Data matching between web data and statistical units is the most challenging issue to solve. Only a good matching will enable to move from classical data collection to online information in the production of JVS. Record Linkage is a promising technique to address the question of matching units. It results from the work of researchers during the last decades who tried to link records of persons and enterprises in large datasets, particularly in the fields of health and demographic studies. The R-RecordLinkage [30] package and the Python recordlinkage toolkit [31] offer a large set of comparison tools including a machine learning approach. Among the many possible string comparisons, the package concentrates on Levensthein and Jarowinkler scores, the second having been found to be the most effective to link addresses, because of its capacity to handle misspellings.

Data matching and deduplication: two facets of a comparison problem

Interestingly, the function “compare” of the LinkageRecord package handles matching and deduplication as two different options of the same issue, with “compare.linkage” resulting in joining cases from two datasets on the basis of similarities of comparison variables, and “compare.dedup” resulting in the exclusion of duplicates identified on the same grounds. Therefore all the methods described in the process of deduplication may potentially apply to matching issues.

Secure long term data acquisition

Secure long term data acquisition is a crucial issue, since public statistics must offer grounds on which to compare changes over time using comparable data. Since Job portals are usually privately developed and are not designed to produce reliable statistics, they are therefore susceptible to significant changes. One must therefore secure a long-term availability of the data, as well as control quality standards over time.

Architectural considerations around OJAs

The current methods in official statistics are based on specific features of traditional statistical data, small amounts of high quality data. These data are collected to fit a specific and predefined statistical purpose and are usually well structured. They are usually collected to ensure a given coverage of population, definitions, methodology and quality in order to meet the needs of official statistics. The guarantee that statistical data are of good quality is the responsibility of the domain expert who designed the survey. This scenario changes when dealing with OJA data. These data can have any kind of structure, they may be structured or semi-structured (data collected from job portals) or completely unstructured (companies' web pages). These data are scraped continuously from the Internet, with dynamic changes of webpages and job portals. The quality of data depends on ability in collecting and pre-processing the data (identifying duplicated adverts, etc.), linked to statistical data (registers), as well as combining with other statistical data sources. These differences between OJA data as opposed to JVS data also force to shift the approach to the analysis: from a model-based approach using a top-down process to a data-driven approach requiring a bottom-up method.

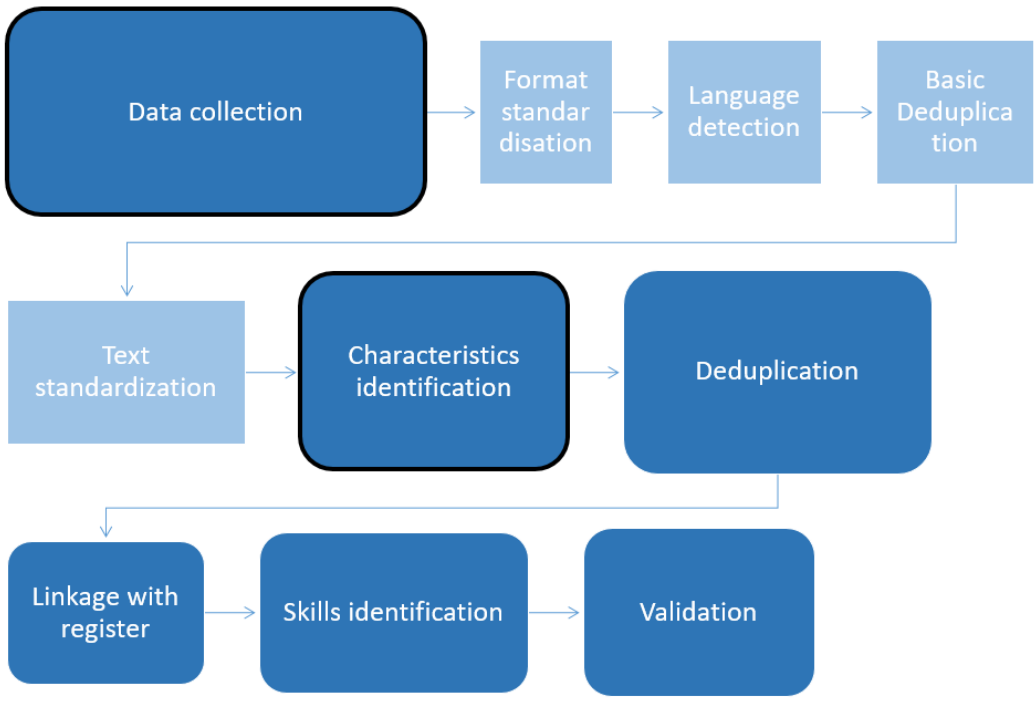


Figure 26: OJA data processing phases

There is a need for developing new models and tools to support the integration of OJA data sources into official statistics. As described in [7] the focus should move to the production of multi-source statistics based on multi-purpose data sources taking into account some principles (including process data at data source):

- Multi-source statistics
- Multi–purpose data sources

- Pushing computation out
- Use data without sharing
- Layered data organisation
- Modular methodological frameworks

An enterprise architecture aims to create an environment which can change and support business goals. It shows what the business needs are, where the organisation wants to be, and ensures that the IT strategy aligns with this. Enterprise architecture helps to remove silos, improves collaboration across an organisation and ensures that the technology is aligned to the business needs. This work also helps standardizing organisations [6].

Models of OJA data integration

With the spread of internet, online job portals have become important places for job matching by employers and employees. Online job portals often collect large number of data in the form of job vacancies and resumes that can serve as a valuable source of information about the characteristics of labour market demand. When selecting OJV portals for gathering information on vacancies it is important to ensure sufficient coverage, especially in countries with a fragmented online job market and where use of such portals is less common. In countries where the online job market is fragmented, more portals need to be scraped to represent the landscape accurately [18]. In order to let alternative data sources gain solid ground in official statistics, [4] propose general framework for integration of these data into the production of official statistics. In the case of using OJA data, three potential scenarios can be foreseen. OJA data need to be pre-processed before they can be linked and integrated to other data sources.

Taking into account the developments and conclusions of successfully concluded ESSnet Big data I [2] integrated approach for collecting OJA data (Figure 27) reduces coverage-related problems. For example Slovenia scrapes main job portals as well as enterprise websites based on a sample for the Job Vacancies Survey (JVS) and thus ensures comparable JVS population and definitions. All sources, including Slovenian employment agency data, are linked with Statistical Business Register and de-duplicated. With the combination of administrative sources, job portal data and enterprise webpages data an estimate of available online job vacancy advertisements present on the reference day or in the reference month are calculated. Furthermore, the deduplication process can be used to estimate the number of newly posted online job vacancies in particular periods.

On the other hand CEDEFOP developed a pan-EU system to gather and analyse OJA data to provide timely information on current and emerging skills needs [16]. CEDEFOP mapped and analysed the OJA landscape in 28 EU Member States to find out how OJAs are used in different countries and portals. To maximise the quality of the information extracted from job adverts, websites were prioritised and ranked by information provided: vacancy release dates; the frequency and regularity of vacancy updates; territorial and sectoral coverage of vacancies; and structured fields in the vacancy notice. To minimise risk of loss of data (unavailability, block, changes in data structure) redundant data sources are ingested. Consequentially effective procedures for identification of noise and duplicate ads have to be implemented to ensure coverage, data quality and definitions.

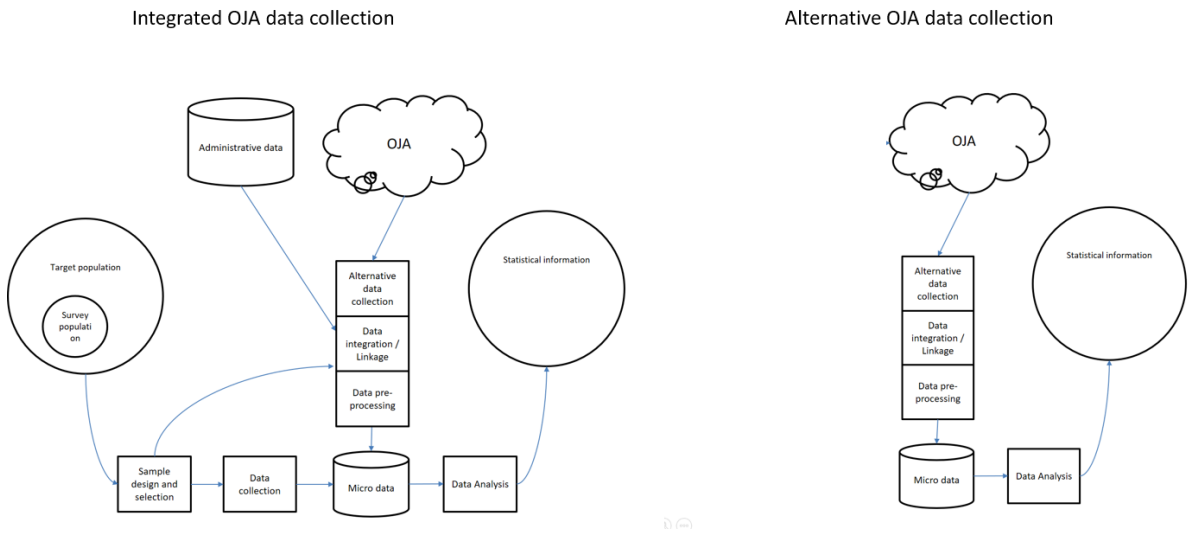


Figure 27: Models of OJA data integration

Both approaches uses similar data processing steps. However they brings important considerations for implementation architecture particularly in the case of common OJA productions system when describing where the used data (OJA data, JVS data, register data) are located (local, European or worldwide level) and how the services (data collection, data processing, information extraction) are autonomous, interoperable, replicated or shared between partners.

OJA Business functions

One approach to describe the business processes for management of OJA data is the Knowledge Discovery in Databases (KDD) process for extracting useful knowledge from unstructured data. The KDD process consists of five main steps, as shown by [3] in Figure 28: data source selection, pre-processing, transformation, data mining and machine learning, interpretation/evaluation.

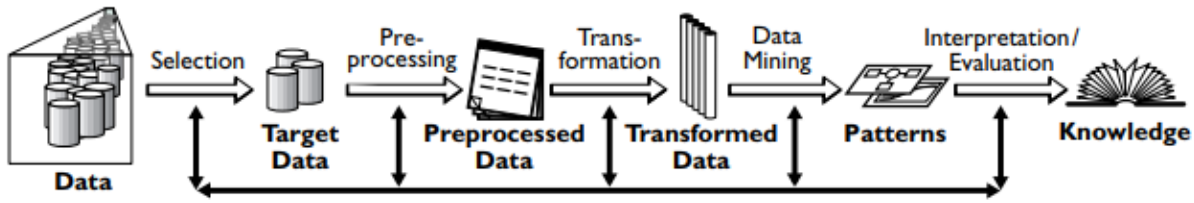


Figure 28: Overview of steps constituting the KDD process

For example according to the KDD CEDEFOP system for processing OJA data is described in five main steps:

1. Data ingestion (formerly known as Crawling): includes all the activities related to data collection. Crawling, fetching, scraping and storing activities are the main tasks of this phase.
2. Pre-processing: includes all the activities related to data preparation for further analysis. Data preparation, translation, data cleaning and text processing tasks are the main activities of this phase.
3. Information extraction: related to the extraction of structured data (occupation, skills, location, industry, etc.) from unstructured texts.

4. Extraction, Transformation and Loading (ETL): concern data preparation for the front-end tools.

While a business process is based on a sequence or flow of activities that is needed to realize a product or service, a business function typically groups behaviour based on required business resources, skills, competencies, knowledge, etc.[8].

To identify related business functions, this process needs to be adapted to the domain of interest, enhancing one task or step with respect to another. In the case of processing OJA data BREAL Big Data REference Architecture and Layers - Business Architecture [8] defines respective business functions.

Table 11: Mapping KDD processes to BREAL Business Functions

KDD process step	BREAL Business Function
Data collection	Acquisition and Recording
Data pre-processing and transformation	Data Wrangling
Data mining and machine learning	Data Representation Modelling and Interpretation
Interpretation/ Evaluation	Shape output Validation

OJA Application services

According to BREAL Big data reference architecture and Layers – Application and Information Architecture [9] each business function consists of several composing functions. For some of them, it is relevant to identify some Application Services, i.e. logical components belonging to a software application layer that implement the specific functions. The BREAL functions from Table 10 for which related application services have been specified are described in the Annex WPB of the same document. Application services presented in Table 12 and detailed described in [9] are described in two levels:

- BREAL Generic Application services
- External Application services

Table 12: BREAL and OJA specific application services

BREAL Business function	BREAL Generic Application services	OJA Specific Application services
Acquisition and recording	External structured data retrieval	Specific scraping, API access
	External unstructured data retrieval	Generic scraping, Website retrieval
	Data storing	Document storing, Relational storing, Key-value storing,
Data wrangling	Data preparation, filtering and deduplication	Data filtering, Language detection, Deduplication and merging
	Data standardization	Format standardisation, Text processing, Data representation and enrichment
	Data aggregation	Pseudo-stock transformation
Data representation	Data derivation	Compute new variables, Entity recognition
	Structure modelling	Data extraction
	Data encoding	Data classification
Modeling and interpretation	Data linking & enriching	Deduplication by linking
	Methodology application	
	Statistical aggregation	
Shape output	Data exchange	

BREAL Generic Application services implements business functions independently from domain of interest whilst External application services implements the same business functions but in OJA specific and detailed way. This approach also support the addition of domain specific application services when identified, for example Pseudo-stock transformation service as shown in Figure 29 with the business functions in yellow and the application services in blue.

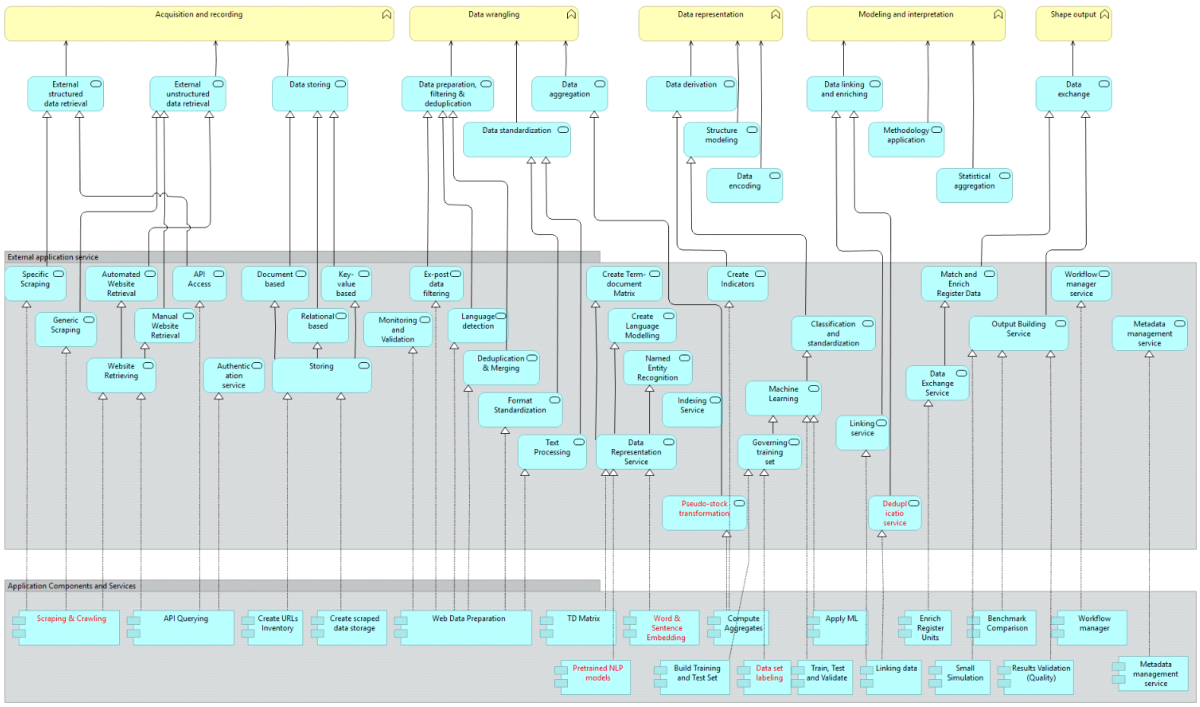


Figure 29: Example of BREAL and OJA specific application services

ESS OJA implementation scenarios

The OJA architecture defines collection OJA data from Internet sources and extraction of useful information from them. The aim is also to decouple the specificity and complexity of OJA data sources from the statistical analysis, following hourglass model [17] and promote collaborated development and sharing of solutions within the ESS. BREAL focuses in particular on the Application Architecture and the Information Architecture, the two architectural layers describing the "how" implementing the "what" defined by the business layer by defining a set of generic application services, proposed with the purpose of showing how the identified business functions could be implemented. In this way the use of BREAL ensure the comparability and shareability of solutions and provides enough details for reusing specific project solutions, through the solution layer.

The ESS EARF defines possible consolidation approaches for the Building Blocks, Services and Information Objects in the SPRA:

- Autonomous
- Interoperable
- Replicated
- Shared

Starting from the interests and building strengths of all parties (Member States and Eurostat), there is a ground for collaborative development of a shared system for producing data from OJA. Such a system should cover the need of all Member States and all official languages gather OJA data from wide range of preselected websites and portals, cover all stages of data processing: data collection, data pre-processing, characteristics identification, information extraction and analysis.

There are various possibilities for NSIs to combine their efforts, working together on a joint solution. The easiest solution is to share code and replicate application services. Another possibility is to share infrastructure and in specific occasions data sources. According to BREAL [8] not all combinations of sharing add value and are not suitable for all countries.

For successful sharing and reusing gradual approach is needed. In first phase country with locally implemented autonomous OJA production system could replicate and test particular service providing country-specific input data and metadata.

In the second phase countries start to proactively participate in the development processes also by provide country-specific input data, metadata and algorithms in order to secure sustainable technical solution for the production of high quality, complete and comparable OJA data. An important aspect of this collaboration is also organisation of work, including possible roles and responsibilities of each party as well as governance structure.

According to SPRA [5], no matter whether a statistical institute chooses to implement one or the other scenario, a gradual implementation remains possible. For these reasons, the suggested approach also takes into account the NSI's specific production systems of those statistical institutes that cannot find positive business cases for moving the production of specific stable statistics to the common production system. According to BREAL in the future, a target should be that the services linked to "Acquisition and recording", "Data wrangling" and "Data representation" will be shared through the NSIs. The most peculiar thing about these three business functions is their autonomy regarding the aim of the process. Whatever the data you want to work on, data acquisition and recording are the same. The data needed could change the parameters used for scraping or website retrieving, but technical background is the same.

OJA functional architecture

Classical NLP (Natural language processing) production lifecycle could be described in four main phases, which continuously evolve due to new data and new features of the data:

- Exploratory data analysis
- Development and testing of ML models and algorithms
- Production
- And continuous improvement

Machine learning is about rapid experimentation and iteration and without keeping track of modelling history; it is not possible to learn much. Effective versioning is needed to keep track of all models developed, as well as their effectiveness, and optimal combination of input and output parameters, algorithm choice and applied architecture. Effective model versioning supports:

- Finding the best model
- Failure tolerance
- Increased complexity and file dependencies
- Gradual, staged deployment

Figure 30 proposes OJA functional architecture, following also some principles of future ESS Web intelligence hub as defined in [7]:

- Serving national and European needs
- Modular structure
- Defined processes and products to be guaranteed
- Priority to working together, possibility to act individually
- Programs should be open source
- Transparency as much as possible
- Commonly used processes should be certified and audible
- Lineage of data and processes
- Intermediate products usable by all partners

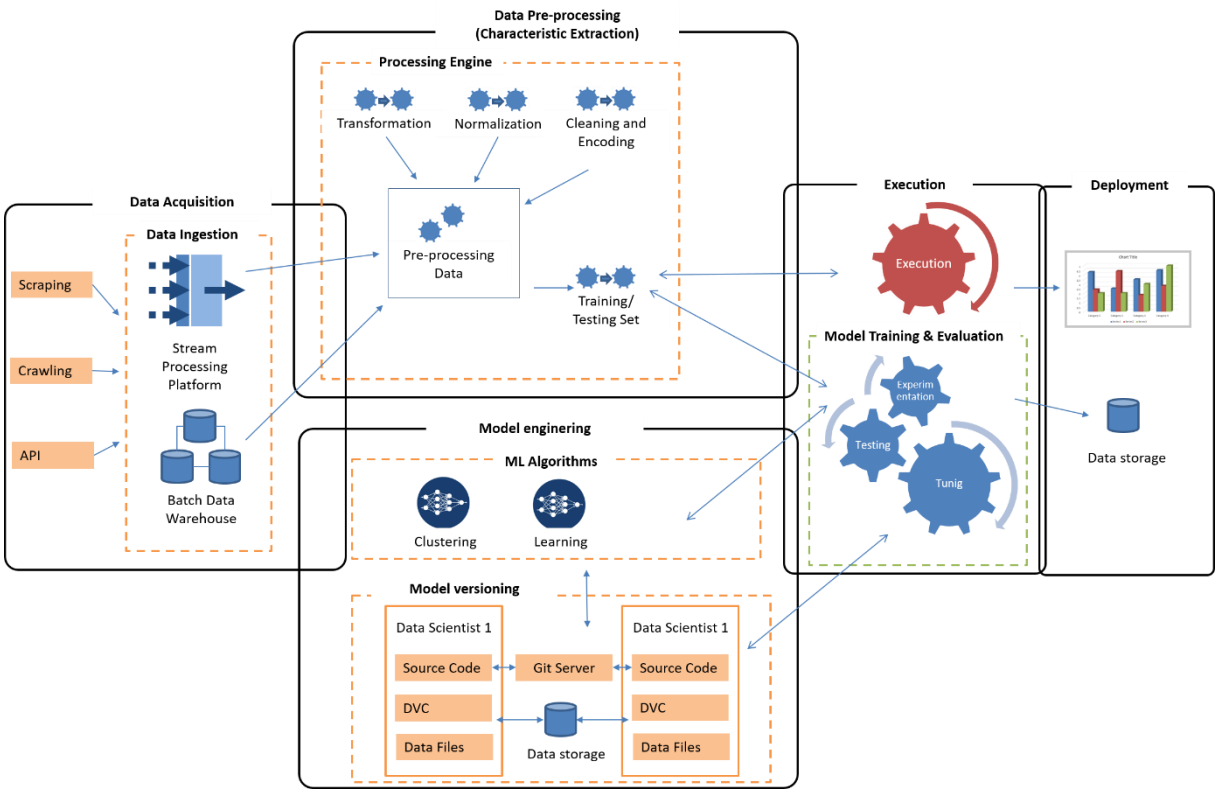


Figure 30: OJA functional architecture

The proposed model could support collaboration in the field of machine learning developments, like management and versioning of datasets and machine learning models making projects reproducible and shareable, helping to answer questions about how a model was built[8].

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